CHAPTER

8 BJT Small-Signal Analysis

8.1 INTRODUCTION

The transistor models introduced in Chapter 7 will now be used to perform a small-signal ac analysis of a number of standard transistor network configurations. The networks analyzed represent the majority of those appearing in practice today. Modifications of the standard configurations will be relatively easy to examine once the content of this chapter is reviewed and understood.

Since the r_e model is sensitive to the actual point of operation, it will be our primary model for the analysis to be performed. For each configuration, however, the effect of an output impedance is examined as provided by the h_{oe} parameter of the hybrid equivalent model. To demonstrate the similarities in analysis that exist between models, a section is devoted to the small-signal analysis of BJT networks using solely the hybrid equivalent model. The analysis of this chapter does not include a load resistance R_L or source resistance R_s. The effect of both parameters is reserved for a systems approach in Chapter 10.

The computer analysis section includes a brief description of the transistor model employed in the PSpice software package. It demonstrates the range and depth of the computer analysis systems available today and how relatively easy it is to enter a complex network and print out the desired results.

8.2 COMMON-EMITTER FIXED-BIAS CONFIGURATION

The first configuration to be analyzed in detail is the common-emitter *fixed-bias* network of Fig. 8.1. Note that the input signal V_i is applied to the base of the transistor while the output V_o is off the collector. In addition, recognize that the input current I_i is not the base current but the source current, while the output current I_o is the collector current. The small-signal ac analysis begins by removing the dc effects of V_{CC} and replacing the dc blocking capacitors C_1 and C_2 by short-circuit equivalents, resulting in the network of Fig. 8.2.

Note in Fig. 8.2 that the common ground of the dc supply and the transistor emitter terminal permits the relocation of R_B and R_C in parallel with the input and output sections of the transistor, respectively. In addition, note the placement of the important network parameters Z_i, Z_o, I_i, and I_o on the redrawn network. Substituting the r_e model for the common-emitter configuration of Fig. 8.2 will result in the network of Fig. 8.3.

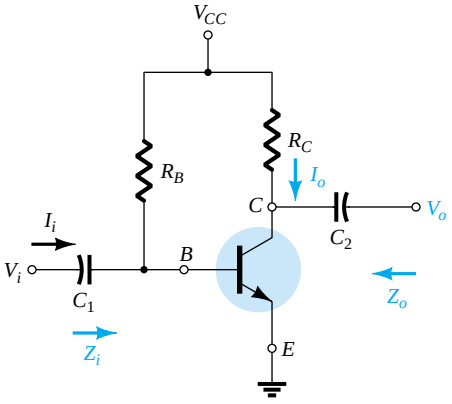


Figure 8.1 Common-emitter fixed-bias configuration.

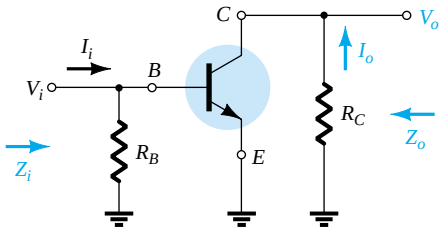


Figure 8.2 Network of Figure 8.1 following the removal of the effects of V_{CC}, C_1, and C_2.

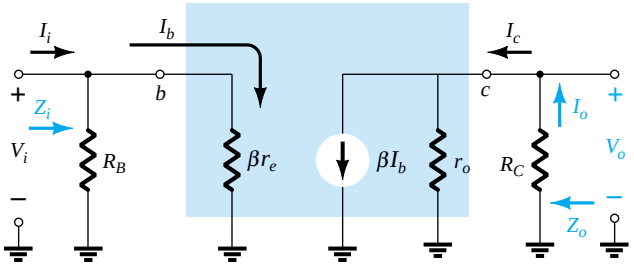


Figure 8.3 Substituting the r_e model into the network of Fig. 8.2.

The next step is to determine β, r_e, and r_o. The magnitude of β is typically obtained from a specification sheet or by direct measurement using a curve tracer or transistor testing instrument. The value of r_e must be determined from a dc analysis of the system, and the magnitude of r_o is typically obtained from the specification sheet or characteristics. Assuming that β, r_e, and r_o have been determined will result in the following equations for the important two-port characteristics of the system.

***Z_i*:** Figure 8.3 clearly reveals that

$$Z_i = R_B \| \beta r_e \quad \text{ohms} \tag{8.1}$$

For the majority of situations R_B is greater than βr_e by more than a factor of 10 (recall from the analysis of parallel elements that the total resistance of two parallel resistors is always less than the smallest and very close to the smallest if one is much larger than the other), permitting the following approximation:

$$Z_i \cong \beta r_e \quad \text{ohms} \tag{8.2}$$
$$R_B \geq 10\beta r_e$$

***Z_o*:** Recall that the output impedance of any system is defined as the impedance Z_o determined when $V_i = 0$. For Fig. 8.3, when $V_i = 0$, $I_i = I_b = 0$, resulting in an open-circuit equivalence for the current source. The result is the configuration of Fig. 8.4.

$$Z_o = R_C \| r_o \quad \text{ohms} \tag{8.3}$$

If $r_o \geq 10\ R_D$, the approximation $R_C \| r_o \cong R_C$ is frequently applied and

$$Z_o \cong R_C \tag{8.4}$$
$$r_o \geq 10R_C$$

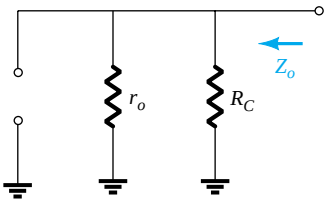
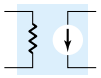


Figure 8.4 Determining Z_o for the network of Fig. 8.3.



A_v: The resistors r_o and R_C are in parallel,

and
$$V_o = -\beta I_b(R_C \| r_o)$$

but
$$I_b = \frac{V_i}{\beta r_c}$$

so that
$$V_o = -\beta\left(\frac{V_i}{\beta_{r_e}}\right)(R_C \| r_o)$$

and
$$\boxed{A_v = \frac{V_o}{V_i} = -\frac{(R_C \| r_o)}{r_e}} \tag{8.5}$$

If $r_o \geq 10R_C$,

$$\boxed{A_v = -\frac{R_C}{r_e}}_{r_o \geq 10R_C} \tag{8.6}$$

Note the explicit absence of β in Eqs. (8.5 and 8.6), although we recognize that β must be utilized to determine r_e.

A_i: The current gain is determined in the following manner: Applying the current-divider rule to the input and output circuits,

$$I_o = \frac{(r_o)(\beta I_b)}{r_o + R_C} \quad \text{and} \quad \frac{I_o}{I_b} = \frac{r_o\beta}{r_o + R_C}$$

with
$$I_b = \frac{(R_B)(I_i)}{R_B + \beta r_e} \quad \text{or} \quad \frac{I_b}{I_i} = \frac{R_B}{R_B + \beta r_e}$$

The result is

$$A_i = \frac{I_o}{I_i} = \left(\frac{I_o}{I_b}\right)\left(\frac{I_b}{I_i}\right) = \left(\frac{r_o\beta}{r_o + R_C}\right)\left(\frac{R_B}{R_B + \beta r_e}\right)$$

and
$$\boxed{A_i = \frac{I_o}{I_i} = \frac{\beta R_B r_o}{(r_o + R_C)(R_B + \beta r_e)}} \tag{8.7}$$

which is certainly an unwieldy, complex expression.

However, if $r_o \geq 10R_C$ and $R_B \geq 10\beta r_e$, which is often the case,

$$A_i = \frac{I_o}{I_i} \cong \frac{\beta R_B r_o}{(r_o)(R_B)}$$

and
$$\boxed{A_i \cong \beta}_{r_o \geq 10R_C,\ R_B \geq 10\beta r_e} \tag{8.8}$$

The complexity of Eq. (8.7) suggests that we may want to return to an equation such as Eq. (7.10), which utilizes A_o and Z_i. That is,

$$\boxed{A_i = -A_v\frac{Z_i}{R_C}} \tag{8.9}$$

Phase Relationship: The negative sign in the resulting equation for A_v reveals that a 180° phase shift occurs between the input and output signals, as shown in Fig. 8.5.

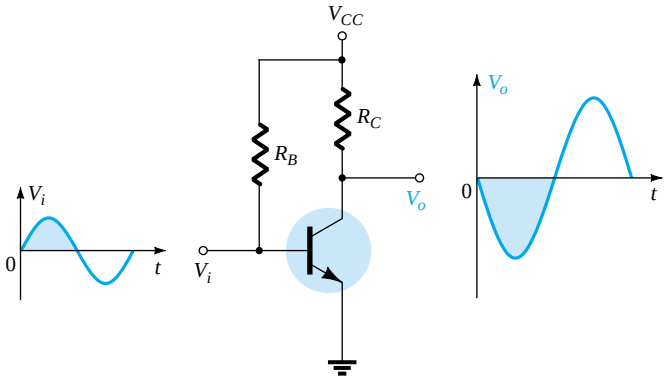


Figure 8.5 Demonstrating the 180° phase shift between input and output waveforms.

EXAMPLE 8.1

For the network of Fig. 8.6:

(a) Determine r_e.
(b) Find Z_i (with $r_o = \infty\ \Omega$).
(c) Calculate Z_o (with $r_o = \infty\ \Omega$).
(d) Determine A_v (with $r_o = \infty\ \Omega$).
(e) Find A_i (with $r_o = \infty\ \Omega$).
(f) Repeat parts (c) through (e) including $r_o = 50\ \text{k}\Omega$ in all calculations and compare results.

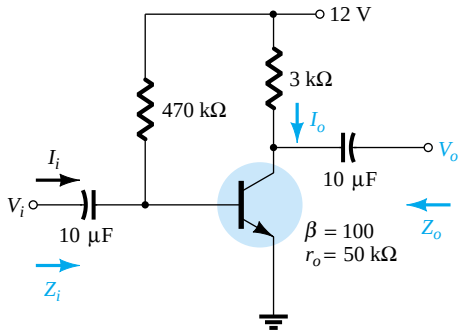


Figure 8.6 Example 8.1.

Solution

(a) DC analysis:

$$I_B = \frac{V_{CC} - V_{BE}}{R_B} = \frac{12\ \text{V} - 0.7\ \text{V}}{470\ \text{k}\Omega} = 24.04\ \mu\text{A}$$

$$I_E = (\beta + 1)I_B = (101)(24.04\ \mu\text{A}) = 2.428\ \text{mA}$$

$$r_e = \frac{26\ \text{mV}}{I_E} = \frac{26\ \text{mV}}{2.428\ \text{mA}} = \mathbf{10.71\ \Omega}$$

(b) $\beta r_e = (100)(10.71\ \Omega) = 1.071\ \text{k}\Omega$

$Z_i = R_B \| \beta r_e = 470\ \text{k}\Omega \| 1.071\ \text{k}\Omega = \mathbf{1.069\ k\Omega}$

(c) $Z_o = R_C = \mathbf{3\ k\Omega}$

(d) $A_v = -\dfrac{R_C}{r_e} = -\dfrac{3\ \text{k}\Omega}{10.71\ \Omega} = \mathbf{-280.11}$

(e) Since $R_B \geq 10\beta r_e (470\ \text{k}\Omega > 10.71\ \text{k}\Omega)$

$A_i \cong \beta = \mathbf{100}$

(f) $Z_o = r_o \| R_C = 50\ \text{k}\Omega \| 3\ \text{k}\Omega = \mathbf{2.83\ k\Omega}$ vs. 3 kΩ

$$A_v = -\frac{r_o \| R_C}{r_e} = \frac{2.83\ \text{k}\Omega}{10.71\ \Omega} = \mathbf{-264.24} \text{ vs. } -280.11$$

$$A_i = \frac{\beta R_B r_o}{(r_o + R_C)(R_B + \beta r_e)} = \frac{(100)(470\ \text{k}\Omega)(50\ \text{k}\Omega)}{(50\ \text{k}\Omega + 3\ \text{k}\Omega)(470\ \text{k}\Omega + 1.071\ \text{k}\Omega)}$$

$$= \mathbf{94.13} \text{ vs. } 100$$

As a check:

$$A_i = -A_v \frac{Z_i}{R_C} = \frac{-(-264.24)(1.069\ \text{k}\Omega)}{3\ \text{k}\Omega} = \mathbf{94.16}$$

which differs slightly only due to the accuracy carried through the calculations.

8.3 VOLTAGE-DIVIDER BIAS

The next configuration to be analyzed is the *voltage-divider* bias network of Fig. 8.7. Recall that the name of the configuration is a result of the voltage-divider bias at the input side to determine the dc level of V_B.

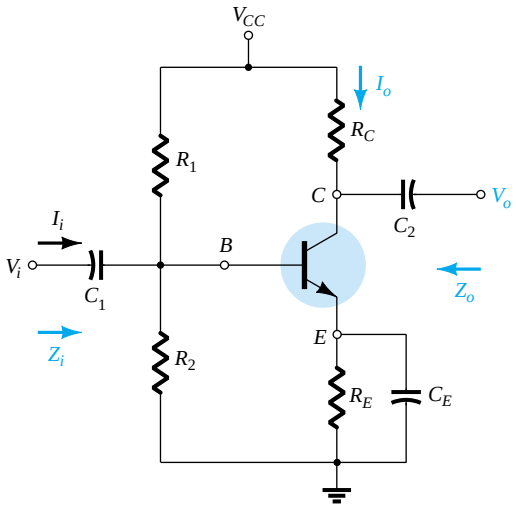


Figure 8.7 Voltage-divider bias configuration.

Substituting the r_e equivalent circuit will result in the network of Fig. 8.8. Note the absence of R_E due to the low-impedance shorting effect of the bypass capacitor, C_E. That is, at the frequency (or frequencies) of operation, the reactance of the capacitor is so small compared to R_E that it is treated as a short circuit across R_E. When

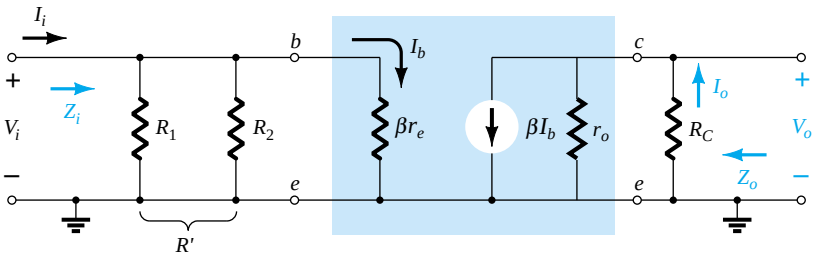


Figure 8.8 Substituting the r_e equivalent circuit into the ac equivalent network of Fig. 8.7.

V_{CC} is set to zero, it places one end of R_1 and R_C at ground potential as shown in Fig. 8.8. In addition, note that R_1 and R_2 remain part of the input circuit while R_C is part of the output circuit. The parallel combination of R_1 and R_2 is defined by

$$R' = R_1 \| R_2 = \frac{R_1 R_2}{R_1 + R_2} \tag{8.10}$$

Z_i: From Fig. 8.8,

$$Z_i = R' \| \beta r_e \tag{8.11}$$

Z_o: From Fig. 8.8 with V_i set to 0 V resulting in $I_b = 0$ μA and $\beta I_b = 0$ mA,

$$Z_o = R_C \| r_o \tag{8.12}$$

If $r_o \geq 10R_C$,

$$Z_o \cong R_C \Big|_{r_o \geq 10R_C} \tag{8.13}$$

A_v: Since R_C and r_o are in parallel,

$$V_o = -(\beta I_b)(R_C \| r_o)$$

and

$$I_b = \frac{V_i}{\beta r_e}$$

so that

$$V_o = -\beta \left(\frac{V_i}{\beta r_e} \right)(R_C \| r_o)$$

and

$$A_v = \frac{V_o}{V_i} = \frac{-R_C \| r_o}{r_e} \tag{8.14}$$

which you will note is an exact duplicate of the equation obtained for the fixed-bias configuration.

For $r_o \geq 10R_C$,

$$A_v = \frac{V_o}{V_i} \cong -\frac{R_C}{r_e} \Big|_{r_o \geq 10R_C} \tag{8.15}$$

A_i: Since the network of Fig. 8.8 is so similar to that of Fig. 8.3 except for the fact that $R' = R_1 \| R_2 = R_B$, the equation for the current gain will have the same format as Eq. (8.7). That is,

$$A_i = \frac{I_o}{I_i} = \frac{\beta R' r_o}{(r_o + R_C)(R' + \beta r_e)} \tag{8.16}$$

For $r_o \geq 10R_C$,

$$A_i = \frac{I_o}{I_i} \cong \frac{\beta R' r_o}{r_o (R' + \beta r_e)}$$

and

$$A_i = \frac{I_o}{I_i} \cong \frac{\beta R'}{R' + \beta r_e} \Big|_{r_o \geq 10R_C} \tag{8.17}$$

And if $R' \geq 10\beta r_e$,

$$A_i = \frac{I_o}{I_i} = \frac{\beta R'}{R'}$$

and

$$\boxed{A_i = \frac{I_o}{I_i} \cong \beta} \quad r_o \geq 10R_C,\ R' \geq 10\beta r_e \tag{8.18}$$

As an option,

$$\boxed{A_i = -A_v \frac{Z_i}{R_C}} \tag{8.19}$$

Phase relationship: The negative sign of Eq. (8.14) reveals a 180° phase shift between V_o and V_i.

EXAMPLE 8.2

For the network of Fig. 8.9, determine:
(a) r_e.
(b) Z_i.
(c) Z_o ($r_o = \infty\ \Omega$).
(d) A_v ($r_o = \infty\ \Omega$).
(e) A_i ($r_o = \infty\ \Omega$).
(f) The parameters of parts (b) through (e) if $r_o = 1/h_{oe} = 50\ \text{k}\Omega$ and compare results.

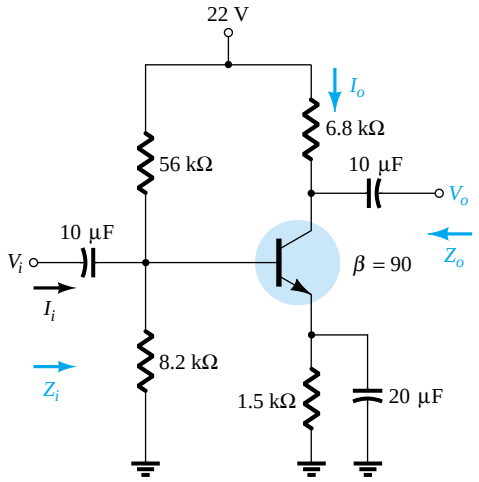


Figure 8.9 Example 8.2.

Solution

(a) DC: Testing $\beta R_E > 10R_2$

$$(90)(1.5\ \text{k}\Omega) > 10(8.2\ \text{k}\Omega)$$

$$135\ \text{k}\Omega > 82\ \text{k}\Omega\ (\textit{satisfied})$$

Using the approximate approach,

$$V_B = \frac{R_2}{R_1 + R_2} V_{CC} = \frac{(8.2\ \text{k}\Omega)(22\ \text{V})}{56\ \text{k}\Omega + 8.2\ \text{k}\Omega} = 2.81\ \text{V}$$

$$V_E = V_B - V_{BE} = 2.81\ \text{V} - 0.7\ \text{V} = 2.11\ \text{V}$$

$$I_E = \frac{V_E}{R_E} = \frac{2.11\ \text{V}}{1.5\ \text{k}\Omega} = 1.41\ \text{mA}$$

$$r_e = \frac{26\ \text{mV}}{I_E} = \frac{26\ \text{mV}}{1.41\ \text{mA}} = \mathbf{18.44\ \Omega}$$

(b) $R' = R_1 \| R_2 = (56\ \text{k}\Omega) \| (8.2\ \text{k}\Omega) = 7.15\ \text{k}\Omega$

$Z_i = R' \| \beta r_e = 7.15\ \text{k}\Omega \| (90)(18.44\ \Omega) = 7.15\ \text{k}\Omega \| 1.66\ \text{k}\Omega$

$= \mathbf{1.35\ k\Omega}$

(c) $Z_o = R_C = \mathbf{6.8\ k\Omega}$

(d) $A_v = -\frac{R_C}{r_e} = -\frac{6.8\ \text{k}\Omega}{18.44\ \Omega} = \mathbf{-368.76}$

(e) The condition $R' \geq 10\beta r_e$ ($7.15\ \text{k}\Omega \geq 10(1.66\ \text{k}\Omega) = 16.6\ \text{k}\Omega$ is *not* satisfied. Therefore,

$$A_i \cong \frac{\beta R'}{R' + \beta r_e} = \frac{(90)(7.15\ \text{k}\Omega)}{7.15\ \text{k}\Omega + 1.66\ \text{k}\Omega} = \mathbf{73.04}$$

(f) $Z_i = \mathbf{1.35\ k\Omega}$

$Z_o = R_C \| r_o = 6.8\ \text{k}\Omega \| 50\ \text{k}\Omega = \mathbf{5.98\ k\Omega}$ vs. $6.8\ \text{k}\Omega$

$A_v = -\frac{R_C \| r_o}{r_e} = -\frac{5.98\ \text{k}\Omega}{18.44\ \Omega} = \mathbf{-324.3}$ vs. -368.76

The condition

$$r_o \geq 10R_C\ (50\ \text{k}\Omega \geq 10(6.8\ \text{k}\Omega) = 68\ \text{k}\Omega)$$

is *not* satisfied. Therefore,

$$A_i = \frac{\beta R' r_o}{(r_o + R_C)(R' + \beta r_e)} = \frac{(90)(7.15\ \text{k}\Omega)(50\ \text{k}\Omega)}{(50\ \text{k}\Omega + 6.8\ \text{k}\Omega)(7.15\ \text{k}\Omega + 1.66\ \text{k}\Omega)}$$

$= \mathbf{64.3}$ vs. 73.04

There was a measurable difference in the results for Z_o, A_v, and A_i because the condition $r_o \geq 10R_C$ was *not* satisfied.

8.4 CE EMITTER-BIAS CONFIGURATION

The networks examined in this section include an emitter resistor that may or may not be bypassed in the ac domain. We will first consider the unbypassed situation and then modify the resulting equations for the bypassed configuration.

Unbypassed

The most fundamental of unbypassed configurations appears in Fig. 8.10. The r_e equivalent model is substituted in Fig. 8.11, but note the absence of the resistance r_o. The effect of r_o is to make the analysis a great deal more complicated, and considering the fact that in most situations its effect can be ignored, it will not be included in the current analysis. However, the effect of r_o will be discussed later in this section.

Applying Kirchhoff's voltage law to the input side of Fig. 8.11 will result in

$$V_i = I_b \beta r_e + I_e R_E$$

or

$$V_i = I_b \beta r_e + (\beta + 1) I_b R_E$$

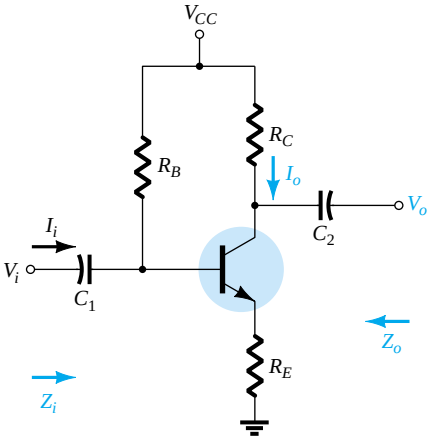


Figure 8.10 CE emitter-bias configuration.

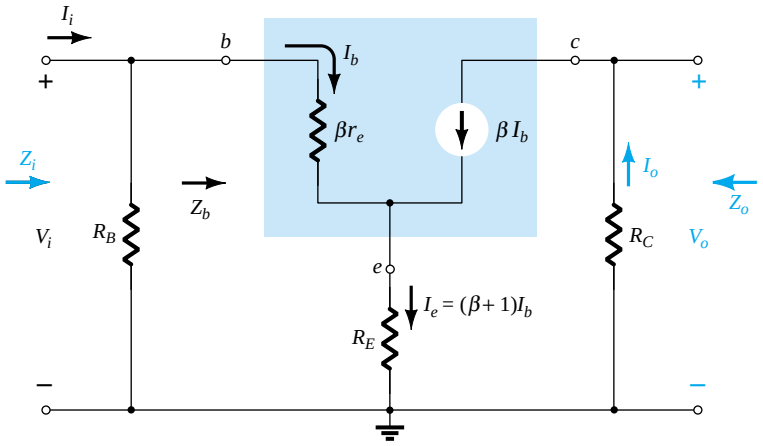


Figure 8.11 Substituting the r_e equivalent circuit into the ac equivalent network of Fig. 8.10.

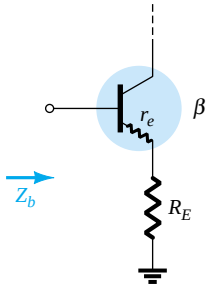


Figure 8.12 Defining the input impedance of a transistor with an unbypassed emitter resistor.

and the input impedance looking into the network to the right of R_B *is*

$$Z_b = \frac{V_i}{I_b} = \beta r_e + (\beta + 1)R_E$$

The result as displayed in Fig. 8.12 reveals that the input impedance of a transistor with an unbypassed resistor R_E is determined by

$$\boxed{Z_b = \beta r_e + (\beta + 1)R_E} \tag{8.20}$$

Since β is normally much greater than 1, the approximate equation is the following:

$$Z_b \cong \beta r_e + \beta R_E$$

and

$$\boxed{Z_b \cong \beta(r_e + R_E)} \tag{8.21}$$

Since R_E is often much greater than r_e, Eq. (8.21) can be further reduced to

$$\boxed{Z_b \cong \beta R_E} \tag{8.22}$$

Z_i: Returning to Fig. 8.11, we have

$$\boxed{Z_i = R_B \| Z_b} \tag{8.23}$$

Z_o: With V_i set to zero, $I_b = 0$ and βI_b can be replaced by an open-circuit equivalent. The result is

$$\boxed{Z_o = R_C} \tag{8.24}$$

A_v:

$$I_b = \frac{V_i}{Z_b}$$

and

$$V_o = -I_o R_C = -\beta I_b R_C$$

$$= -\beta\left(\frac{V_i}{Z_b}\right) R_C$$

with

$$\boxed{A_v = \frac{V_o}{V_i} = -\frac{\beta R_C}{Z_b}} \tag{8.25}$$

Substituting $Z_b = \beta(r_e + R_E)$ gives

$$\boxed{A_v = \frac{V_o}{V_i} = -\frac{R_C}{r_e + R_E}} \tag{8.26}$$

and for the approximation $Z_b \cong \beta R_E$,

$$\boxed{A_v = \frac{V_o}{V_i} \cong -\frac{R_C}{R_E}} \tag{8.27}$$

Note again the absence of β from the equation for A_v.

A_i: The magnitude of R_B is often too close to Z_b to permit the approximation $I_b = I_i$. Applying the current-divider rule to the input circuit will result in

$$I_b = \frac{R_B I_i}{R_B + Z_b}$$

and

$$\frac{I_b}{I_i} = \frac{R_B}{R_B + Z_b}$$

In addition,

$$I_o = \beta I_b$$

and

$$\frac{I_o}{I_b} = \beta$$

so that

$$A_i = \frac{I_o}{I_i} = \frac{I_o}{I_b}\frac{I_b}{I_i}$$

$$= \beta \frac{R_B}{R_B + Z_b}$$

and

$$\boxed{A_i = \frac{I_o}{I_i} = \frac{\beta R_B}{R_B + Z_b}} \tag{8.28}$$

or

$$\boxed{A_i = -A_v \frac{Z_i}{R_C}} \tag{8.29}$$

Phase relationship: The negative sign in Eq. (8.25) again reveals a 180° phase shift between V_o and V_i.

Effect of r_o: The equations appearing below will clearly reveal the additional complexity resulting from including r_o in the analysis. Note in each case, however, that when certain conditions are met, the equations return to the form just derived. The derivation of each equation is beyond the needs of this text and is left as an exercise for the reader. Each equation can be derived through *careful* application of the basic laws of circuit analysis such as Kirchhoff's voltage and current laws, source conversions, Thévenin's theorem, and so on. The equations were included to remove the nagging question of the effect of r_o on the important parameters of a transistor configuration.

Z_i:

$$\boxed{Z_b = \beta r_e + \left[\frac{(\beta + 1) + R_C/r_o}{1 + (R_C + R_E)/r_o}\right] R_E} \tag{8.30}$$

Since the ratio R_C/r_o is always much less than $(\beta + 1)$,

$$Z_b \cong \beta r_e + \frac{(\beta + 1)R_E}{1 + (R_C + R_E)/r_o}$$

For $r_o \geq 10(R_C + R_E)$,

$$Z_b \cong \beta r_e + (\beta + 1)R_E$$

which compares directly with Eq. (8.20).

In other words, if $r_o \geq 10(R_C + R_E)$, all the equations derived earlier will result.

Since $\beta + 1 \cong \beta$, the following equation is an excellent one for most applications:

$$\boxed{Z_b \cong \beta(r_e + R_E)}_{r_o \geq 10(R_C + R_E)} \tag{8.31}$$

Z_o:

$$\boxed{Z_o = R_C \| \left[r_o + \frac{\beta(r_o + r_e)}{1 + \dfrac{\beta r_e}{R_E}} \right]} \tag{8.32}$$

However, $r_o \gg r_e$, and

$$Z_o \cong R_C \| r_o \left[1 + \frac{\beta}{1 + \dfrac{\beta r_e}{R_E}} \right]$$

which can be written as

$$Z_o \cong R_C \| r_o \left[1 + \frac{1}{\dfrac{1}{\beta} + \dfrac{r_e}{R_E}} \right]$$

Typically $1/\beta$ and r_e/R_E are less than one with a sum usually less than one. The result is a multiplying factor for r_o greater than one. For $\beta = 100$, $r_e = 10\ \Omega$, and $R_E = 1\ \text{k}\Omega$:

$$\frac{1}{\dfrac{1}{\beta} + \dfrac{r_e}{R_E}} = \frac{1}{\dfrac{1}{100} + \dfrac{10\ \Omega}{1000\ \Omega}} = \frac{1}{0.02} = 50$$

and

$$Z_o = R_C \| 51 r_o$$

which is certainly simply R_C. Therefore,

$$\boxed{Z_o = R_C}_{\text{Any level of } r_o} \tag{8.33}$$

which was obtained earlier.

A_v:

$$\boxed{A_v = \frac{V_o}{V_i} = \frac{-\dfrac{\beta R_C}{Z_b}\left[1 + \dfrac{r_e}{r_o}\right] + \dfrac{R_C}{r_o}}{1 + \dfrac{R_C}{r_o}}} \tag{8.34}$$

The ratio $\frac{r_e}{r_o} \ll 1$

and
$$A_v = \frac{V_o}{V_i} \cong \frac{-\frac{\beta R_C}{Z_b} + \frac{R_C}{r_o}}{1 + \frac{R_C}{r_o}}$$

For $r_o \geq 10R_C$,

$$\boxed{A_v = \frac{V_o}{V_i} \cong -\frac{\beta R_C}{Z_b}} \quad r_o \geq 10R_C \tag{8.35}$$

as obtained earlier.

A_i: The determination of A_i will be left to the equation

$$\boxed{A_i = -A_v \frac{Z_i}{R_C}} \tag{8.36}$$

using the above equations.

Bypassed

If R_E of Fig. 8.10 is bypassed by an emitter capacitor C_E, the complete r_e equivalent model can be substituted resulting in the same equivalent network as Fig. 8.3. Eqs. (8.1 through 8.9) are therefore applicable.

EXAMPLE 8.3

For the network of Fig. 8.13, without C_E (unbypassed), determine:
(a) r_e.
(b) Z_i.
(c) Z_o.
(d) A_v.
(e) A_i.

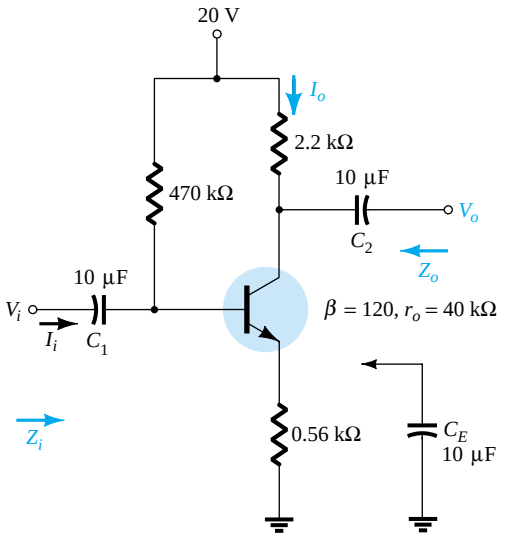


Figure 8.13 Example 8.3.

Solution

(a) DC: $I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E} = \frac{20\text{ V} - 0.7\text{ V}}{470\text{ k}\Omega + (121)0.56\text{ k}\Omega} = 35.89\ \mu\text{A}$

$I_E = (\beta + 1)I_B = (121)(46.5\ \mu\text{A}) = 4.34\text{ mA}$

and $r_e = \frac{26\text{ mV}}{I_E} = \frac{26\text{ mV}}{4.34\text{ mA}} = \mathbf{5.99\ \Omega}$

(b) Testing the condition $r_o \geq 10(R_C + R_E)$,

$$40\ \text{k}\Omega \geq 10(2.2\ \text{k}\Omega + 0.56\ \text{k}\Omega)$$

$$40\ \text{k}\Omega \geq 10(2.76\ \text{k}\Omega) = 27.6\ \text{k}\Omega\ (\textit{satisfied})$$

Therefore,

$$Z_b \cong \beta(r_e + R_E) = 120(5.99\ \Omega + 560\ \Omega)$$
$$= 67.92\ \text{k}\Omega$$

and

$$Z_i = R_B \| Z_b = 470\ \text{k}\Omega \| 67.92\ \text{k}\Omega$$
$$= \mathbf{59.34\ k\Omega}$$

(c) $Z_o = R_C = \mathbf{2.2\ k\Omega}$

(d) $r_o \geq 10R_C$ is satisfied. Therefore,

$$A_v = \frac{V_o}{V_i} \cong -\frac{\beta R_C}{Z_b} = -\frac{(120)(2.2\ \text{k}\Omega)}{67.92\ \text{k}\Omega}$$
$$= \mathbf{-3.89}$$

compared to -3.93 using Eq. (8.27): $A_v \cong -R_C/R_E$.

(e) $$A_i = -A_v \frac{Z_i}{R_C} = -(-3.89)\left(\frac{59.34\ \text{k}\Omega}{2.2\ \text{k}\Omega}\right)$$
$$= \mathbf{104.92}$$

compared to 104.85 using Eq. (8.28): $A_i \cong \beta R_B/(R_B + Z_b)$.

EXAMPLE 8.4 Repeat the analysis of Example 8.3 with C_E in place.

Solution

(a) The dc analysis is the same, and $r_e = 5.99\ \Omega$.

(b) R_E is "shorted out" by C_E for the ac analysis. Therefore,

$$Z_i = R_B \| Z_b = R_B \| \beta r_e = 470\ \text{k}\Omega \| (120)(5.99\ \Omega)$$
$$= 470\ \text{k}\Omega \| 718.8\ \Omega \cong \mathbf{717.70\ \Omega}$$

(c) $Z_o = R_C = \mathbf{2.2\ k\Omega}$

(d) $$A_v = -\frac{R_C}{r_e}$$
$$= -\frac{2.2\ \text{k}\Omega}{5.99\ \Omega} = \mathbf{-367.28}$$ (a significant increase)

(e) $$A_i = \frac{\beta R_B}{R_B + Z_b} = \frac{(120)(470\ \text{k}\Omega)}{470\ \text{k}\Omega + 718.8\ \Omega}$$
$$= \mathbf{119.82}$$

EXAMPLE 8.5 For the network of Fig. 8.14, determine (using appropriate approximations):

(a) r_e.

(b) Z_i.

(c) Z_o.

(d) A_v.

(e) A_i.

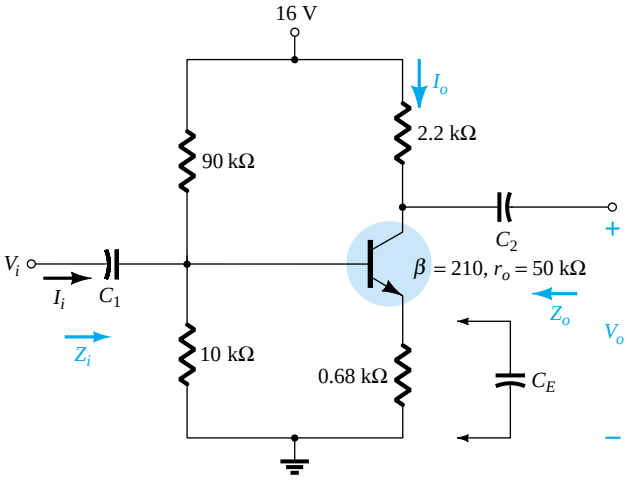


Figure 8.14 Example 8.5.

Solution

(a) Testing $\beta R_E > 10R_2$

$$(210)(0.68\ \text{k}\Omega) > 10(10\ \text{k}\Omega)$$

$$142.8\ \text{k}\Omega > 100\ \text{k}\Omega\ (\textit{satisfied})$$

$$V_B = \frac{R_2}{R_1 + R_2} V_{CC} = \frac{10\ \text{k}\Omega}{90\ \text{k}\Omega + 10\ \text{k}\Omega}(16\ \text{V}) = 1.6\ \text{V}$$

$$V_E = V_B - V_{BE} = 1.6\ \text{V} - 0.7\ \text{V} = 0.9\ \text{V}$$

$$I_E = \frac{V_E}{R_E} = \frac{0.9\ \text{V}}{0.68\ \text{k}\Omega} = 1.324\ \text{mA}$$

$$r_e = \frac{26\ \text{mV}}{I_E} = \frac{26\ \text{mV}}{1.324\ \text{mA}} = \mathbf{19.64\ \Omega}$$

(b) The ac equivalent circuit is provided in Fig. 8.15. The resulting configuration is now different from Fig. 8.11 only by the fact that now

$$R_B = R' = R_1 \| R_2 = 9\ \text{k}\Omega$$

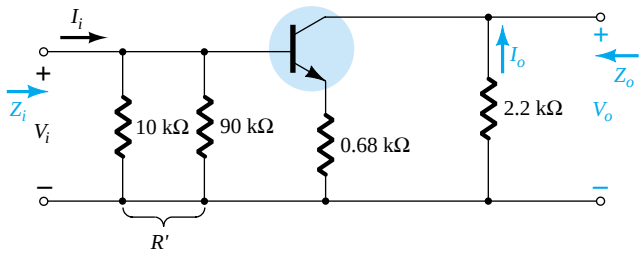


Figure 8.15 The ac equivalent circuit of Fig. 8.14.

The testing conditions of $r_o \geq 10(R_C + R_E)$ and $r_o \geq 10R_C$ are both satisfied. Using the appropriate approximations yields

$$Z_b \cong \beta R_E = 142.8\ \text{k}\Omega$$

$$Z_i = R_B \| Z_b = 9\ \text{k}\Omega \| 142.8\ \text{k}\Omega$$

$$= \mathbf{8.47\ k\Omega}$$

(c) $Z_o = R_C = \mathbf{2.2\ k\Omega}$

(d) $A_v = -\dfrac{R_C}{R_E} = -\dfrac{2.2\ \text{k}\Omega}{0.68\ \text{k}\Omega} = \mathbf{-3.24}$

(e) $A_i = -A_v\dfrac{Z_i}{R_C} = -(-3.24)\left(\dfrac{8.47\ \text{k}\Omega}{2.2\ \text{k}\Omega}\right)$

$= \mathbf{12.47}$

EXAMPLE 8.6 Repeat Example 8.5 with C_E in place.

Solution

(a) The dc analysis is the same, and $r_e = \mathbf{19.64\ \Omega}$.

(b) $Z_b = \beta r_e = (210)(19.64\ \Omega) \cong 4.12\ \text{k}\Omega$

$Z_i = R_B \| Z_b = 9\ \text{k}\Omega \| 4.12\ \text{k}\Omega$

$= \mathbf{2.83\ k\Omega}$

(c) $Z_o = R_C = \mathbf{2.2\ k\Omega}$

(d) $A_v = -\dfrac{R_C}{r_e} = -\dfrac{2.2\ \text{k}\Omega}{19.64\ \text{k}\Omega} = \mathbf{-112.02}$ (a significant increase)

(e) $A_i = -A_v\dfrac{Z_i}{R_L} = -(-112.02)\left(\dfrac{2.83\ \text{k}\Omega}{2.2\ \text{k}\Omega}\right)$

$= \mathbf{144.1}$

Another variation of an emitter-bias configuration appears in Fig. 8.16. For the dc analysis, the emitter resistance is $R_{E_1} + R_{E_2}$, while for the ac analysis, the resistor R_E in the equations above is simply R_{E_1} with R_{E_2} bypassed by C_E.

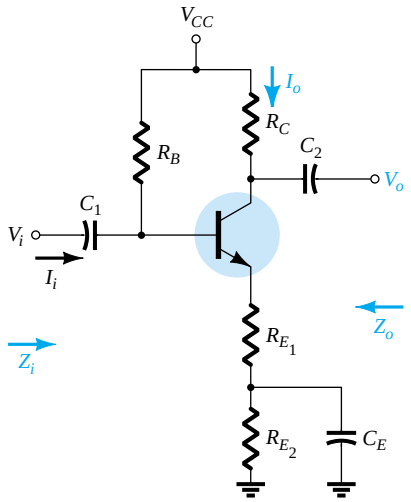


Figure 8.16 An emitter-bias configuration with a portion of the emitter-bias resistance bypassed in the ac domain.

8.5 EMITTER-FOLLOWER CONFIGURATION

When the output is taken from the emitter terminal of the transistor as shown in Fig. 8.17, the network is referred to as an *emitter-follower.* The output voltage is always slightly less than the input signal due to the drop from base to emitter, but the ap-

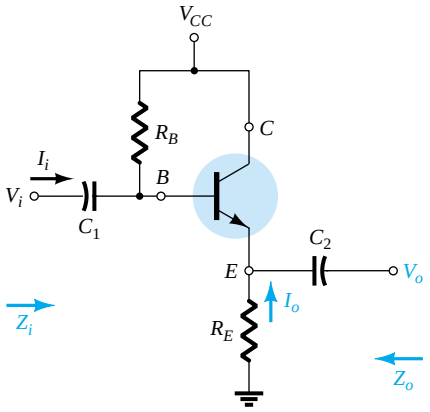


Figure 8.17 Emitter-follower configuration.

proximation $A_v \cong 1$ is usually a good one. Unlike the collector voltage, the emitter voltage is in phase with the signal V_i. That is, both V_o and V_i will attain their positive and negative peak values at the same time. The fact that V_o "follows" the magnitude of V_i with an in-phase relationship accounts for the terminology emitter-follower.

The most common emitter-follower configuration appears in Fig. 8.17. In fact, because the collector is grounded for ac analysis, it is actually a *common-collector* configuration. Other variations of Fig. 8.17 that draw the output off the emitter with $V_o \cong V_i$ will appear later in this section.

The emitter-follower configuration is frequently used for impedance-matching purposes. It presents a high impedance at the input and a low impedance at the output, which is the direct opposite of the standard fixed-bias configuration. The resulting effect is much the same as that obtained with a transformer, where a load is matched to the source impedance for maximum power transfer through the system.

Substituting the r_e equivalent circuit into the network of Fig. 8.17 will result in the network of Fig. 8.18. The effect of r_o will be examined later in the section.

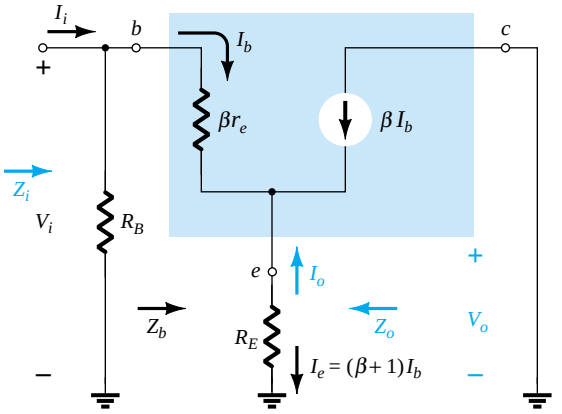


Figure 8.18 Substituting the r_e equivalent circuit into the ac equivalent network of Fig. 8.17.

Z_i: The input impedance is determined in the same manner as described in the preceding section:

$$Z_i = R_B \| Z_b \tag{8.37}$$

with

$$Z_b = \beta r_e + (\beta + 1)R_E \tag{8.38}$$

or $$Z_b \cong \beta(r_e + R_E) \tag{8.39}$$

and $$Z_b \cong \beta R_E \tag{8.40}$$

Z_o: The output impedance is best described by first writing the equation for the current I_b:

$$I_b = \frac{V_i}{Z_b}$$

and then multiplying by $(\beta + 1)$ to establish I_e. That is,

$$I_e = (\beta + 1)I_b = (\beta + 1)\frac{V_i}{Z_b}$$

Substituting for Z_b gives

$$I_e = \frac{(\beta + 1)V_i}{\beta r_e + (\beta + 1)R_E}$$

or $$I_e = \frac{V_i}{[\beta r_e/(\beta + 1)] + R_E}$$

but $$(\beta + 1) \cong \beta$$

and $$\frac{\beta r_e}{\beta + 1} \cong \frac{\beta r_e}{\beta} = r_e$$

so that $$I_e \cong \frac{V_i}{r_e + R_E} \tag{8.41}$$

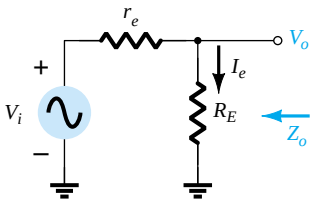


Figure 8.19 Defining the output impedance for the emitter-follower configuration.

If we now construct the network defined by Eq. (8.41), the configuration of Fig. 8.19 will result.

To determine Z_o, V_i is set to zero and

$$Z_o = R_E \| r_e \tag{8.42}$$

Since R_E is typically much greater than r_e, the following approximation is often applied:

$$Z_o \cong r_e \tag{8.43}$$

A_v: Figure 8.19 can be utilized to determine the voltage gain through an application of the voltage-divider rule:

$$V_o = \frac{R_E V_i}{R_E + r_e}$$

and $$A_v = \frac{V_o}{V_i} = \frac{R_E}{R_E + r_e} \tag{8.44}$$

Since R_E is usually much greater than r_e, $R_E + r_e \cong R_E$ and

$$A_v = \frac{V_o}{V_i} \cong 1 \tag{8.45}$$

A_i: From Fig. 8.18,

$$I_b = \frac{R_B I_i}{R_B + Z_b}$$

or

$$\frac{I_b}{I_i} = \frac{R_B}{R_B + Z_b}$$

and

$$I_o = -I_e = -(\beta + 1)I_b$$

or

$$\frac{I_o}{I_b} = -(\beta + 1)$$

so that

$$A_i = \frac{I_o}{I_i} = \frac{I_o}{I_b}\frac{I_b}{I_i}$$

$$= -(\beta + 1)\frac{R_B}{R_B + Z_b}$$

and since

$$(\beta + 1) \cong \beta,$$

$$\boxed{A_i \cong -\frac{\beta R_B}{R_B + Z_b}} \tag{8.46}$$

or

$$\boxed{A_i = -A_v \frac{Z_i}{R_E}} \tag{8.47}$$

Phase relationship: As revealed by Eq. (8.44) and earlier discussions of this section, V_o and V_i are in phase for the emitter-follower configuration.

Effect of r_o:

Z_i:

$$\boxed{Z_b = \beta r_e + \frac{(\beta + 1)R_E}{1 + \dfrac{R_E}{r_o}}} \tag{8.48}$$

If the condition $r_o \geq 10R_E$ is satisfied,

$$Z_b = \beta r_e + (\beta + 1)R_E$$

which matches earlier conclusions with

$$\boxed{Z_b \cong \beta(r_e + R_E)}_{r_o \geq 10R_E} \tag{8.49}$$

Z_o:

$$\boxed{Z_o = r_o \| R_E \| \frac{\beta r_e}{(\beta + 1)}} \tag{8.50}$$

Using $\beta + 1 \cong \beta$,

$$Z_o = r_o \| R_E \| r_e$$

and since $r_o \gg r_e$,

$$\boxed{Z_o \cong R_E \| r_e}_{\text{Any } r_o} \tag{8.51}$$

A_v:

$$A_v = \frac{(\beta + 1)R_E/Z_b}{1 + \frac{R_E}{r_o}} \tag{8.52}$$

If the condition $r_o \geq 10R_E$ is satisfied and we use the approximation $\beta + 1 \cong \beta$,

$$A_v \cong \frac{\beta R_E}{Z_b}$$

But $$Z_b \cong \beta(r_e + R_E)$$

so that $$A_v \cong \frac{\beta R_E}{\beta(r_e + R_E)}$$

and

$$A_v \cong \frac{R_E}{r_e + R_E}\Big|_{r_o \geq 10R_E} \tag{8.53}$$

EXAMPLE 8.7

For the emitter-follower network of Fig. 8.20, determine:
(a) r_e.
(b) Z_i.
(c) Z_o.
(d) A_v.
(e) A_i.
(f) Repeat parts (b) through (e) with $r_o = 25\ \text{k}\Omega$ and compare results.

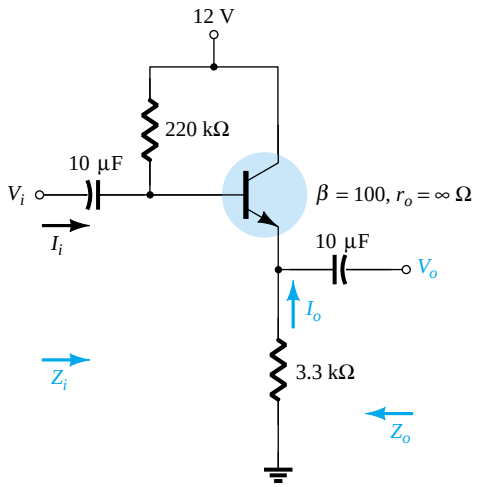


Figure 8.20 Example 8.7.

Solution

(a) $$I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E}$$

$$= \frac{12\ \text{V} - 0.7\ \text{V}}{220\ \text{k}\Omega + (101)3.3\ \text{k}\Omega} = 20.42\ \mu\text{A}$$

$$I_E = (\beta + 1)I_B$$

$$= (101)(20.42\ \mu\text{A}) = 2.062\ \text{mA}$$

$$r_e = \frac{26\ \text{mV}}{I_E} = \frac{26\ \text{mV}}{2.062\ \text{mA}} = \mathbf{12.61\ \Omega}$$

(b) $Z_b = \beta r_e + (\beta + 1)R_E$

$= (100)(12.61\ \Omega) + (101)(3.3\ \text{k}\Omega)$

$= 1.261\ \text{k}\Omega + 333.3\ \text{k}\Omega$

$= 334.56\ \text{k}\Omega \cong \beta R_E$

$Z_i = R_B \| Z_b = 220\ \text{k}\Omega \| 334.56\ \text{k}\Omega$

$= \mathbf{132.72\ k\Omega}$

(c) $Z_o = R_E \| r_e = 3.3\ \text{k}\Omega \| 12.61\ \Omega$

$\mathbf{= 12.56\ \Omega} \cong r_e$

(d) $A_v = \dfrac{V_o}{V_i} = \dfrac{R_E}{R_E + r_e} = \dfrac{3.3\ \text{k}\Omega}{3.3\ \text{k}\Omega + 12.61\ \Omega}$

$= \mathbf{0.996} \cong 1$

(e) $A_i \cong -\dfrac{\beta R_B}{R_B + Z_b} = -\dfrac{(100)(220\ \text{k}\Omega)}{220\ \text{k}\Omega + 334.56\ \text{k}\Omega} = \mathbf{-39.67}$

versus

$$A_i = -A_v \frac{Z_i}{R_E} = -(0.996)\left(\frac{132.72\ \text{k}\Omega}{3.3\ \text{k}\Omega}\right) = \mathbf{-40.06}$$

(f) Checking the condition $r_o \geq 10R_E$, we have

$$25\ \text{k}\Omega \geq 10(3.3\ \text{k}\Omega) = 33\ \text{k}\Omega$$

which is *not* satisfied. Therefore,

$$Z_b = \beta r_e + \frac{(\beta + 1)R_E}{1 + \dfrac{R_E}{r_o}} = (100)(12.61\ \Omega) + \frac{(100 + 1)3.3\ \text{k}\Omega}{1 + \dfrac{3.3\ \text{k}\Omega}{25\ \text{k}\Omega}}$$

$= 1.261\ \text{k}\Omega + 294.43\ \text{k}\Omega$

$= 295.7\ \text{k}\Omega$

with $Z_i = R_B \| Z_b = 220\ \text{k}\Omega \| 295.7\ \text{k}\Omega$

$= \mathbf{126.15\ k\Omega}$ vs. $132.72\ \text{k}\Omega$ obtained earlier

$Z_o = R_E \| r_e = \mathbf{12.56\ \Omega}$ as obtained earlier

$$A_v = \frac{(\beta + 1)R_E / Z_b}{\left[1 + \dfrac{R_E}{r_o}\right]} = \frac{(100 + 1)(3.3\ \text{k}\Omega)/295.7\ \text{k}\Omega}{\left[1 + \dfrac{3.3\ \text{k}\Omega}{25\ \text{k}\Omega}\right]}$$

$= \mathbf{0.996} \cong 1$

matching the earlier result.

In general, therefore, even though the condition $r_o \geq 10R_E$ was not satisfied, the results for Z_o and A_v are the same, with Z_i only slightly less. The results suggest that for most applications a good approximation for the actual results can be obtained by simply ignoring the effects of r_o for this configuration.

The network of Fig. 8.21 is a variation of the network of Fig. 8.17, which employs a voltage-divider input section to set the bias conditions. Equations (8.37) through (8.47) are changed only by replacing R_B by $R' = R_1 \| R_2$.

The network of Fig. 8.22 will also provide the input/output characteristics of an emitter-follower but includes a collector resistor R_C. In this case R_B is again replaced by the parallel combination of R_1 and R_2. The input impedance Z_i and output impedance Z_o are unaffected by R_C since it is not reflected into the base or emitter equiv-

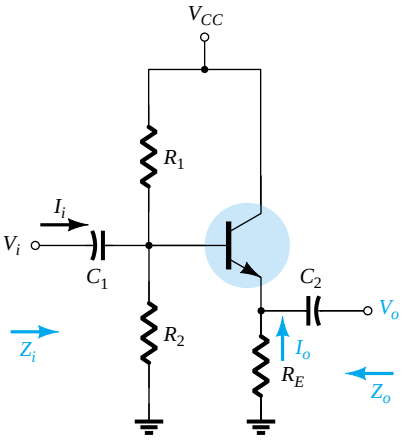


Figure 8.21 Emitter-follower configuration with a voltage-divider biasing arrangement.

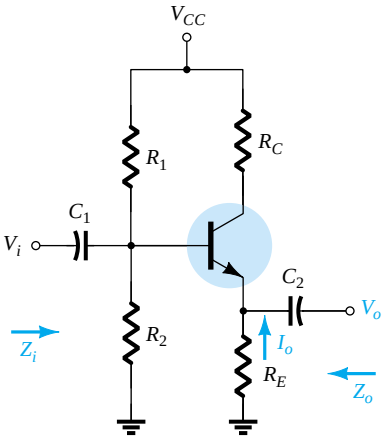


Figure 8.22 Emitter-follower configuration with a collector resistor R_C.

alent networks. In fact, the only effect of R_C will be to determine the Q-point of operation.

8.6 COMMON-BASE CONFIGURATION

The common-base configuration is characterized as having a relatively low input and a high output impedance and a current gain less than 1. The voltage gain, however, can be quite large. The standard configuration appears in Fig. 8.23, with the common-base r_e equivalent model substituted in Fig. 8.24. The transistor output impedance r_o is not included for the common-base configuration because it is typically in the megohm range and can be ignored in parallel with the resistor R_C.

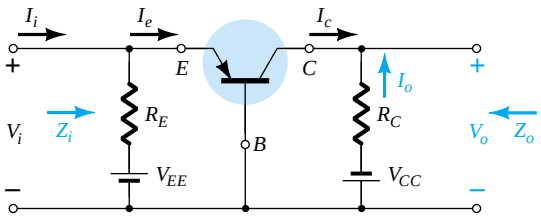


Figure 8.23 Common-base configuration.

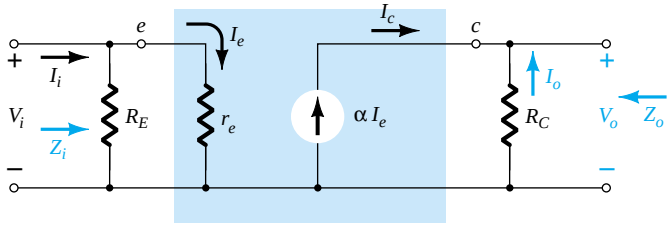


Figure 8.24 Substituting the r_e equivalent circuit into the ac equivalent network of Fig. 8.23.

Z_i:

$$Z_i = R_E \| r_e \tag{8.54}$$

Z_o:

$$Z_o = R_C \tag{8.55}$$

A_v:

$$V_o = -I_o R_C = -(-I_c)R_C = \alpha I_e R_C$$

with $$I_e = \frac{V_i}{r_e}$$

so that $$V_o = \alpha\left(\frac{V_i}{r_e}\right) R_C$$

and $$\boxed{A_v = \frac{V_o}{V_i} = \frac{\alpha R_C}{r_e} \cong \frac{R_C}{r_e}} \qquad (8.56)$$

***A_i*:** Assuming that $R_E \gg r_e$ yields

$$I_e = I_i$$

and $$I_o = -\alpha I_e = -\alpha I_i$$

with $$\boxed{A_i = \frac{I_o}{I_i} = -\alpha \cong -1} \qquad (8.57)$$

Phase relationship: The fact that A_v is a positive number reveals that V_o and V_i are in phase for the common-base configuration.

Effect of r_o: For the common-base configuration, $r_o = 1/h_{ob}$ is typically in the megohm range and sufficiently larger than the parallel resistance R_C to permit the approximation $r_o \| R_C \cong R_C$.

EXAMPLE 8.8

For the network of Fig. 8.25, determine:
(a) r_e.
(b) Z_i.
(c) Z_o.
(d) A_v.
(e) A_i.

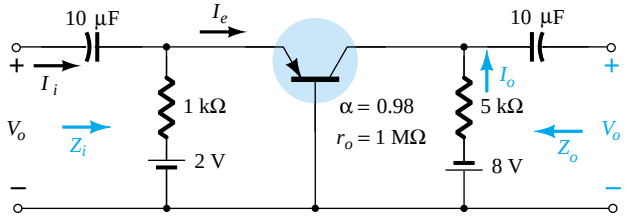


Figure 8.25 Example 8.8.

Solution

(a) $$I_E = \frac{V_{EE} - V_{BE}}{R_E} = \frac{2\text{ V} - 0.7\text{ V}}{1\text{ k}\Omega} = \frac{1.3\text{ V}}{1\text{ k}\Omega} = 1.3\text{ mA}$$

$$r_e = \frac{26\text{ mV}}{I_E} = \frac{26\text{ mV}}{1.3\text{ mA}} = \mathbf{20\ \Omega}$$

(b) $$Z_i = R_E \| r_e = 1\text{ k}\Omega \| 20\ \Omega$$
$$= \mathbf{19.61\ \Omega} \cong r_e$$

(c) $$Z_o = R_C = \mathbf{5\ k\Omega}$$

(d) $$A_v \cong \frac{R_C}{r_e} = \frac{5\text{ k}\Omega}{20\ \Omega} = \mathbf{250}$$

(e) $$A_i = \mathbf{-0.98} \cong -1$$

8.7 COLLECTOR FEEDBACK CONFIGURATION

The collector feedback network of Fig. 8.26 employs a feedback path from collector to base to increase the stability of the system as discussed in Section 4.12. However, the simple maneuver of connecting a resistor from base to collector rather than base to dc supply has a significant impact on the level of difficulty encountered when analyzing the network.

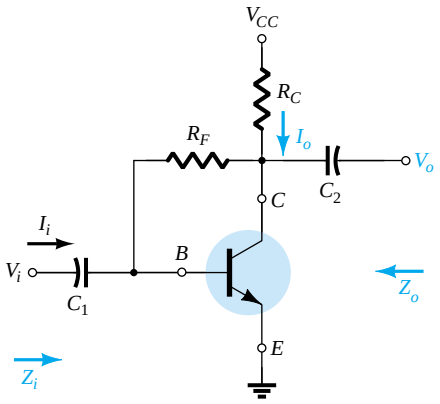


Figure 8.26 Collector feedback configuration.

Some of the steps to be performed below are the result of experience working with such configurations. It is not expected that a new student of the subject would choose the sequence of steps described below without taking a wrong step or two. Substituting the equivalent circuit and redrawing the network will result in the configuration of Fig. 8.27. The effects of a transistor output resistance r_o will be discussed later in the section.

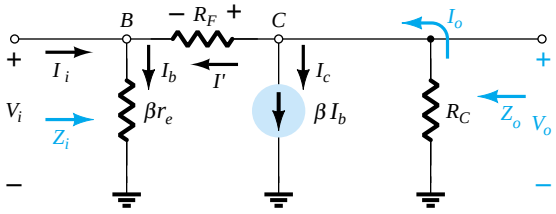


Figure 8.27 Substituting the r_e equivalent circuit into the ac equivalent network of Fig. 8.26.

$\boldsymbol{Z_i}$:
$$I' = \frac{V_o - V_i}{R_F}$$

with
$$V_o = -I_o R_C$$

and
$$I_o = \beta I_b + I'$$

Since βI_b is normally much larger than I',

$$I_o \cong \beta I_b$$

and
$$V_o = -(\beta I_b)R_C = -\beta I_b R_C$$

but
$$I_b = \frac{V_i}{\beta r_e}$$

and
$$V_o = -\beta\left(\frac{V_i}{\beta r_e}\right)R_C = -\frac{R_C}{r_e}V_i$$

Therefore,

$$I' = \frac{V_o - V_i}{R_F} = \frac{V_o}{R_F} - \frac{V_i}{R_F} = -\frac{R_C V_i}{r_e R_F} - \frac{V_i}{R_F} = -\frac{1}{R_F}\left[1 + \frac{R_C}{r_e}\right]V_i$$

The result is

$$V_i = I_b \beta r_e = (I_i + I')\beta r_e = I_i \beta r_e + I' \beta r_e$$

$$V_i = I_i \beta r_e - \frac{1}{R_F}\left[1 + \frac{R_C}{r_e}\right]\beta r_e V_i$$

or

$$V_i\left[1 + \frac{\beta r_e}{R_F}\left[1 + \frac{R_C}{r_e}\right]\right] = I_i \beta r_e$$

and

$$Z_i = \frac{V_i}{I_i} = \frac{\beta r_e}{1 + \dfrac{\beta r_e}{R_F}\left[1 + \dfrac{R_C}{r_e}\right]}$$

but R_C is usually much greater than r_e and $1 + \dfrac{R_C}{r_e} \cong \dfrac{R_C}{r_e}$

so that

$$Z_i = \frac{\beta r_e}{1 + \dfrac{\beta R_C}{R_F}}$$

or

$$Z_i = \frac{r_e}{\dfrac{1}{\beta} + \dfrac{R_C}{R_F}} \tag{8.58}$$

Z_o: If we set V_i to zero as required to define Z_o, the network will appear as shown in Fig. 8.28. The effect of βr_e is removed and R_F appears in parallel with R_C and

$$Z_o \cong R_C \| R_F \tag{8.59}$$

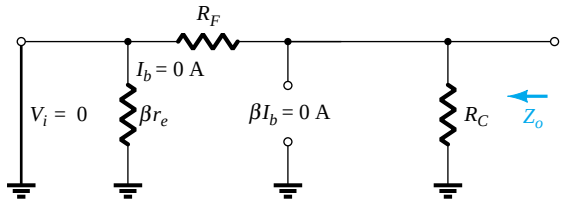


Figure 8.28 Defining Z_o for the collector feedback configuration.

A_v: At node C of Fig. 8.27,

$$I_o = \beta I_b + I'$$

For typical values, $\beta I_b \gg I'$ and $I_o \cong \beta I_b$.

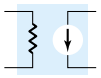
$$V_o = -I_o R_C = -(\beta I_b) R_C$$

Substituting $I_b = V_i/\beta r_e$ gives us

$$V_o = -\beta \frac{V_i}{\beta r_e} R_C$$

and

$$A_v = \frac{V_o}{V_i} = -\frac{R_C}{r_e} \tag{8.60}$$



A_i: Applying Kirchhoff's voltage law around the outside network loop yields

$$V_i + V_{R_F} - V_o = 0$$

and

$$I_b\beta r_e + (I_b - I_i)R_F + I_oR_C = 0$$

Using $I_o \cong \beta I_b$, we have

$$I_b\beta r_e + I_bR_F - I_iR_F + \beta I_bR_C = 0$$

and

$$I_b(\beta r_e + R_F + \beta R_C) = I_iR_F$$

Substituting $I_b = I_o/\beta$ from $I_o \cong \beta I_b$ yields

$$\frac{I_o}{\beta}(\beta r_e + R_F + \beta R_C) = I_iR_F$$

and

$$I_o = \frac{\beta R_F I_i}{\beta r_e + R_F + \beta R_C}$$

Ignoring βr_e compared to R_F and βR_C gives us

$$\boxed{A_i = \frac{I_o}{I_i} = \frac{\beta R_F}{R_F + \beta R_C}} \tag{8.61}$$

For $\beta R_C \gg R_F$,

$$A_i = \frac{I_o}{I_i} = \frac{\beta R_F}{\beta R_C}$$

and

$$\boxed{A_i = \frac{I_o}{I_i} \cong \frac{R_F}{R_C}} \tag{8.62}$$

Phase relationship: The negative sign of Eq. (8.60) reveals a 180° phase shift between V_o and V_i.

Effect of r_o:

Z_i: A complete analysis without applying approximations will result in

$$\boxed{Z_i = \frac{1 + \dfrac{R_C \| r_o}{R_F}}{\dfrac{1}{\beta r_e} + \dfrac{1}{R_F} + \dfrac{R_C \| r_o}{R_F r_e}}} \tag{8.63}$$

Recognizing that $1/R_F \cong 0$ and applying the condition $r_o \geq 10R_C$,

$$Z_i = \frac{1 + \dfrac{R_C}{R_F}}{\dfrac{1}{\beta r_e} + \dfrac{R_C}{R_F r_e}}$$

but typically $R_C/R_F \ll 1$ and

$$Z_i = \frac{1}{\dfrac{1}{\beta r_e} + \dfrac{R_C}{R_F r_e}}$$

or
$$\boxed{Z_i = \frac{r_e}{\frac{1}{\beta} + \frac{R_C}{R_F}}}_{r_o \geq 10R_C} \tag{8.64}$$

as obtained earlier.

Z_o: Including r_o in parallel with R_C in Fig. 8.28 will result in

$$\boxed{Z_o = r_o \| R_C \| R_F} \tag{8.65}$$

For $r_o \geq 10R_C$,

$$\boxed{Z_o \cong R_C \| R_F}_{r_o \geq 10R_C} \tag{8.66}$$

as obtained earlier. For the common condition of $R_F \gg R_C$,

$$\boxed{Z_o \cong R_C}_{r_o \geq 10R_C,\ R_F \gg R_C} \tag{8.67}$$

A_v:

$$\boxed{A_v = -\frac{\left[\frac{1}{R_F} + \frac{1}{r_e}\right](r_o \| R_C)}{1 + \frac{r_o \| R_C}{R_F}}} \tag{8.68}$$

Since $R_F \gg r_e$,

$$A_v \cong -\frac{\frac{r_o \| R_C}{r_e}}{1 + \frac{r_o \| R_C}{R_F}}$$

For $r_o \geq 10R_C$,

$$\boxed{A_v \cong -\frac{\frac{R_C}{r_e}}{1 + \frac{R_C}{R_F}}}_{r_o \geq 10R_C} \tag{8.69}$$

and since R_C/R_F is typically much less than one,

$$\boxed{A_v \cong -\frac{R_C}{r_e}}_{r_o \geq 10R_C,\ R_F \gg R_C} \tag{8.70}$$

as obtained earlier.

EXAMPLE 8.9

For the network of Fig. 8.29, determine:
(a) r_e.
(b) Z_i.
(c) Z_o.
(d) A_v.
(e) A_i.
(f) Repeat parts (b) through (e) with $r_o = 20\ \text{k}\Omega$ and compare results.

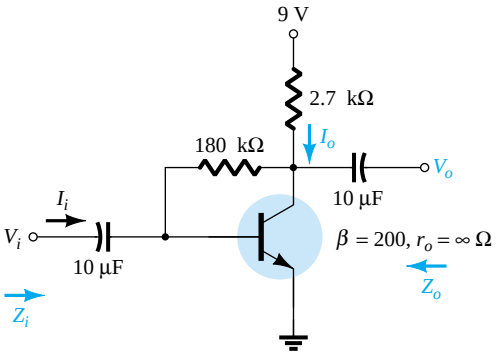


Figure 8.29 Example 8.9.

Solution

(a) $$I_B = \frac{V_{CC} - V_{BE}}{R_F + \beta R_C} = \frac{9\text{ V} - 0.7\text{ V}}{180\text{ k}\Omega + (200)2.7\text{ k}\Omega}$$

$$= 11.53\ \mu\text{A}$$

$$I_E = (\beta + 1)I_B = (201)(11.53\ \mu\text{A}) = 2.32\text{ mA}$$

$$r_e = \frac{26\text{ mV}}{I_E} = \frac{26\text{ mV}}{2.32\text{ mA}} = \mathbf{11.21\ \Omega}$$

(b) $$Z_i = \frac{r_e}{\frac{1}{\beta} + \frac{R_C}{R_F}} = \frac{11.21\ \Omega}{\frac{1}{200} + \frac{2.7\text{ k}\Omega}{180\text{ k}\Omega}} = \frac{11.21\ \Omega}{0.005 + 0.015}$$

$$= \frac{11.21\ \Omega}{0.02} = 50(11.21\ \Omega) = \mathbf{560.5\ \Omega}$$

(c) $$Z_o = R_C \| R_F = 2.7\text{ k}\Omega \| 180\text{ k}\Omega = \mathbf{2.66\ k\Omega}$$

(d) $$A_v = -\frac{R_C}{r_e} = -\frac{27\text{ k}\Omega}{11.21\ \Omega} = \mathbf{-240.86}$$

(e) $$A_i = \frac{\beta R_F}{R_F + \beta R_C} = \frac{(200)(180\text{ k}\Omega)}{180\text{ k}\Omega + (200)(2.7\text{ k}\Omega)}$$

$$= \mathbf{50}$$

(f) Z_i: The condition $r_o \geq 10R_C$ is *not* satisfied. Therefore,

$$Z_i = \frac{1 + \frac{R_C \| r_o}{R_F}}{\frac{1}{\beta r_e} + \frac{1}{R_F} + \frac{R_C \| r_o}{R_F r_e}} = \frac{1 + \frac{2.7\text{ k}\Omega \| 20\text{ k}\Omega}{180\text{ k}\Omega}}{\frac{1}{(200)(11.21)} + \frac{1}{180\text{ k}\Omega} + \frac{2.7\text{ k}\Omega \| 20\text{ k}\Omega}{(180\text{ k}\Omega)(11.21\ \Omega)}}$$

$$= \frac{1 + \frac{2.38\text{ k}\Omega}{180\text{ k}\Omega}}{0.45 \times 10^{-3} + 0.006 \times 10^{-3} + 1.18 \times 10^{-3}} = \frac{1 + 0.013}{1.64 \times 10^{-3}}$$

$$= \mathbf{617.7\ \Omega}\text{ vs. } 560.5\ \Omega \text{ above}$$

Z_o:

$$Z_o = r_o \| R_C \| R_F = 20\text{ k}\Omega \| 2.7\text{ k}\Omega \| 180\text{ k}\Omega$$

$$= \mathbf{2.35\ k\Omega}\text{ vs. } 2.66\text{ k}\Omega \text{ above}$$

A_v:

$$A_v = \frac{-\left[\frac{1}{R_F} + \frac{1}{r_e}\right](r_o \| R_C)}{1 + \frac{r_o \| R_C}{R_F}} = \frac{-\left[\frac{1}{180\ \text{k}\Omega} + \frac{1}{11.21\Omega}\right](2.38\ \text{k}\Omega)}{1 + \frac{2.38\ \text{k}\Omega}{180\ \text{k}\Omega}}$$

$$= \frac{-[5.56 \times 10^{-6} - 8.92 \times 10^{-2}](2.38\ \text{k}\Omega)}{1 + 0.013}$$

$$= \mathbf{-209.56} \text{ vs. } -240.86 \text{ above}$$

A_i:

$$A_i = -A_v \frac{Z_i}{R_C}$$

$$= -(-209.56)\frac{617.7\ \Omega}{2.7\ \text{k}\Omega}$$

$$= \mathbf{47.94} \text{ vs. } 50 \text{ above}$$

For the configuration of Fig. 8.30, Eqs. (8.71) through (8.74) will determine the variables of interest. The derivations are left as an exercise at the end of the chapter.

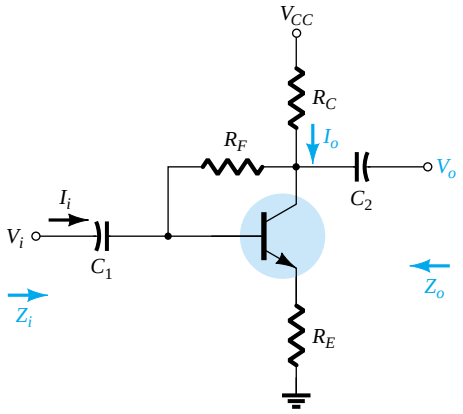


Figure 8.30 Collector feedback configuration with an emitter resistor R_E.

$\boldsymbol{Z_i}$:

$$Z_i \cong \frac{R_E}{\left[\frac{1}{\beta} + \frac{(R_E + R_C)}{R_F}\right]} \tag{8.71}$$

$\boldsymbol{Z_o}$:

$$Z_o \cong R_C \| R_F \tag{8.72}$$

$\boldsymbol{A_v}$:

$$A_v \cong -\frac{R_C}{R_E} \tag{8.73}$$

$\boldsymbol{A_i}$:

$$A_i \cong \frac{1}{\frac{1}{\beta} + \frac{(R_E + R_C)}{R_F}} \tag{8.74}$$

8.8 COLLECTOR DC FEEDBACK CONFIGURATION

The network of Fig. 8.31 has a dc feedback resistor for increased stability, yet the capacitor C_3 will shift portions of the feedback resistance to the input and output sections of the network in the ac domain. The portion of R_F shifted to the input or output side will be determined by the desired ac input and output resistance levels.

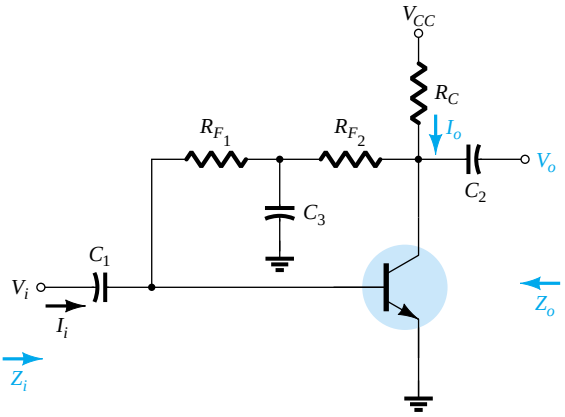


Figure 8.31 Collector dc feedback configuration.

At the frequency or frequencies of operation, the capacitor will assume a short-circuit equivalent to ground due to its low impedance level compared to the other elements of the network. The small-signal ac equivalent circuit will then appear as shown in Fig. 8.32.

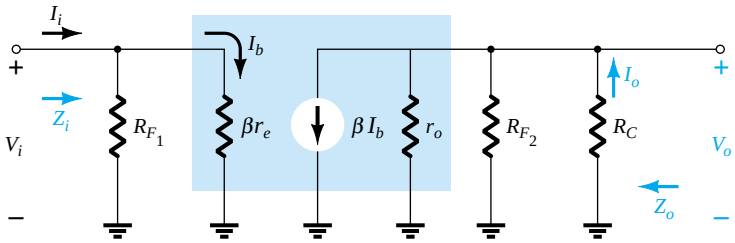


Figure 8.32 Substituting the r_e equivalent circuit into the ac equivalent network of Fig. 8.31.

Z_i:

$$Z_i = R_{F_1} \| \beta r_e \tag{8.75}$$

Z_o:

$$Z_o = R_C \| R_{F_2} \| r_o \tag{8.76}$$

For $r_o \geq 10R_C$,

$$Z_o \cong R_C \| R_{F_2} \Big|_{r_o \geq 10R_C} \tag{8.77}$$

A_v:

$$R' = r_o \| R_{F_2} \| R_C$$

and

$$V_o = -\beta I_b R'$$

but $$I_b = \frac{V_i}{\beta r_e}$$

and $$V_o = -\beta \frac{V_i}{\beta r_e} R'$$

so that

$$\boxed{A_v = \frac{V_o}{V_i} = -\frac{r_o \| R_{F_2} \| R_C}{r_e}} \tag{8.78}$$

For $r_o \geq 10R_C$,

$$\boxed{A_v = \frac{V_o}{V_i} \cong -\frac{R_{F_2} \| R_C}{r_e}}_{r_o \geq 10R_C} \tag{8.79}$$

A_i: For the input side,

$$I_b = \frac{R_{F_1} I_i}{R_{F_1} + \beta r_e} \qquad \text{or} \qquad \frac{I_b}{I_i} = \frac{R_{F_1}}{R_{F_1} + \beta r_e}$$

and for the output side using $R' = r_o \| R_{F_2}$

$$I_o = \frac{R' \beta I_b}{R' + R_C} \qquad \text{or} \qquad \frac{I_o}{I_b} = \frac{R' \beta}{R' + R_C}$$

The current gain,

$$A_i = \frac{I_o}{I_i} = \frac{I_o}{I_b} \cdot \frac{I_b}{I_i}$$
$$= \frac{R' \beta}{R' + R_C} \cdot \frac{R_{F_1}}{R_{F_1} + \beta r_e}$$

and $$\boxed{A_i = \frac{I_o}{I_i} = \frac{\beta R_{F_1} R'}{(R_{F_1} + \beta r_e)(R' + R_C)}}_{R' = r_o \| R_{F_2}} \tag{8.80}$$

Since R_{F_1} is usually much larger than βr_e, $R_{F_1} + \beta r_e \cong R_{F_1}$

and $$A_i = \frac{I_o}{I_i} \cong \frac{\beta R_{F_1} (r_o \| R_{F_2})}{R_{F_1} (r_o \| R_{F_2} + R_C)}$$

so that $$\boxed{A_i = \frac{I_o}{I_i} \cong \frac{\beta}{1 + \dfrac{R_C}{r_o \| R_{F_2}}}}_{R_{F_1} \geq 10\beta r_e} \tag{8.81}$$

or $$\boxed{A_i = \frac{I_o}{I_i} = -A_v \frac{Z_i}{R_C}} \tag{8.82}$$

Phase relationship: The negative sign in Eq. (8.78) clearly reveals a 180° phase shift between input and output voltages.

EXAMPLE 8.10

For the network of Fig. 8.33, determine:

(a) r_e.
(b) Z_i.
(c) Z_o.
(d) A_v.
(e) A_i.

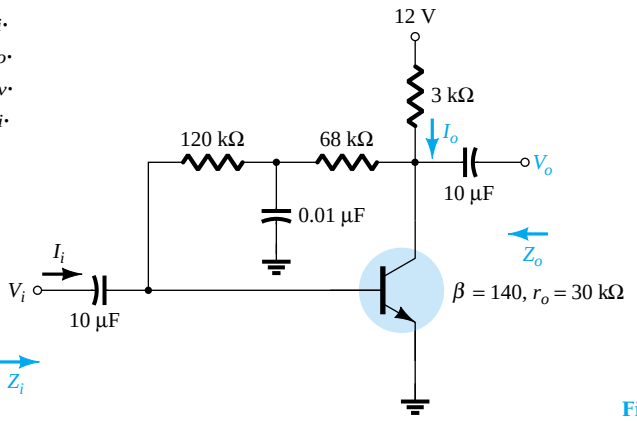


Figure 8.33 Example 8.10.

Solution

(a) DC: $$I_B = \frac{V_{CC} - V_{BE}}{R_F + \beta R_C}$$

$$= \frac{12\text{ V} - 0.7\text{ V}}{(120\text{ k}\Omega + 68\text{ k}\Omega) + (140)3\text{ k}\Omega}$$

$$= \frac{11.3\text{V}}{608\text{ k}\Omega} = 18.6\mu\text{A}$$

$$I_E = (\beta + 1)I_B = (141)(18.6\mu\text{A})$$

$$= 2.62\text{ mA}$$

$$r_e = \frac{26\text{ mV}}{I_E} = \frac{26\text{ mV}}{2.62\text{ mA}} = \mathbf{9.92\ \Omega}$$

(b) $\beta r_e = (140)(9.92\ \Omega) = 1.39\text{ k}\Omega$

The ac equivalent network appears in Fig. 8.34.

$$Z_i = R_{F_1} \| \beta r_e = 120\text{ k}\Omega \| 1.39\text{ k}\Omega$$

$$\cong \mathbf{1.37\ k\Omega}$$

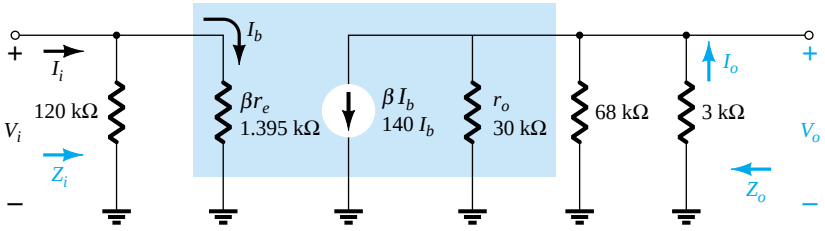


Figure 8.34 Substituting the r_e equivalent circuit into the ac equivalent network of Fig. 8.33.

(c) Testing the condition $r_o \geq 10R_C$, we find

$$30\text{ k}\Omega \geq 10(3\text{ k}\Omega) = 30\text{ k}\Omega$$

which is satisfied through the equals sign in the condition. Therefore,

$$Z_o \cong R_C \| R_{F_2} = 3\text{ k}\Omega \| 68\text{ k}\Omega$$

$$= \mathbf{2.87\ k\Omega}$$

(d) $r_o \geq 10R_C$, therefore,

$$A_v \cong -\frac{R_{F_2} \| R_C}{r_e} = -\frac{68\ \text{k}\Omega \| 3\ \text{k}\Omega}{9.92\ \Omega}$$

$$\cong -\frac{2.87\ \text{k}\Omega}{9.92\ \Omega}$$

$$\cong \mathbf{-289.3}$$

(e) Since the condition $R_{F_1} \gg \beta r_e$ is satisfied,

$$A_i \cong \frac{\beta}{1 + \dfrac{R_C}{r_o \| R_{F_2}}} = \frac{140}{1 + \dfrac{3\ \text{k}\Omega}{30\ \text{k}\Omega \| 68\ \text{k}\Omega}} = \frac{140}{1 + 0.14} = \frac{140}{1.14}$$

$$\cong \mathbf{122.8}$$

8.9 APPROXIMATE HYBRID EQUIVALENT CIRCUIT

The analysis using the approximate hybrid equivalent circuit of Fig. 8.35 for the common-emitter configuration and of Fig. 8.36 for the common-base configuration is very similar to that just performed using the r_e model. Although time and priorities do not permit a detailed analysis of all the configurations discussed thus far, a brief overview of some of the most important will be included in this section to demonstrate the similarities in approach and the resulting equations.

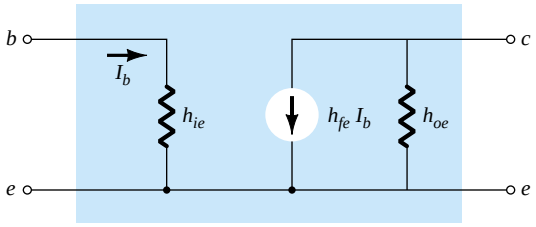


Figure 8.35 Approximate common-emitter hybrid equivalent circuit.

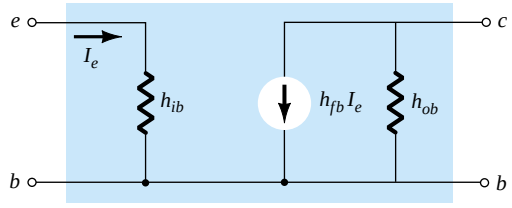


Figure 8.36 Approximate common-base hybrid equivalent circuit.

Since the various parameters of the hybrid model are specified by a data sheet or experimental analysis, the dc analysis associated with use of the r_e model is not an integral part of the use of the hybrid parameters. In other words, when the problem is presented, the parameters such as h_{ie}, h_{fe}, h_{ib}, and so on, are specified. Keep in mind, however, that the hybrid parameters and components of the r_e model are related by the following equations as discussed in detail in Chapter 7: $h_{ie} = \beta r_e$, $h_{fe} = \beta$, $h_{oe} = 1/r_o$, $h_{fb} = -\alpha$, and $h_{ib} = r_e$ (note Appendix A).

Fixed-Bias Configuration

For the fixed-bias configuration of Fig. 8.37, the small-signal ac equivalent network will appear as shown in Fig. 8.38 using the approximate common-emitter hybrid equiv-

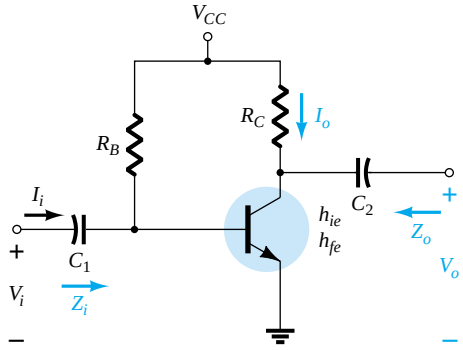


Figure 8.37 Fixed-bias configuration.

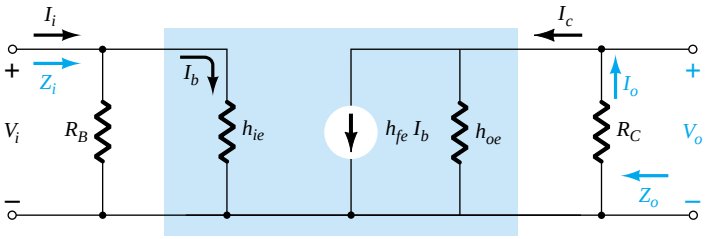


Figure 8.38 Substituting the approximate hybrid equivalent circuit into the ac equivalent network of Fig. 8.37.

alent model. Compare the similarities in appearance with Fig. 8.3 and the r_e model analysis. The similarities suggest that the analysis will be quite similar, and the results of one can be directly related to the other.

Z_i: From Fig. 8.38,

$$\boxed{Z_i = R_B \| h_{ie}} \tag{8.83}$$

Z_o: From Fig. 8.38,

$$\boxed{Z_o = R_C \| 1/h_{oe}} \tag{8.84}$$

A_v: Using $R' = 1/h_{oe} \| R_C$,

$$V_o = -I_o R' = -I_C R'$$
$$= -h_{fe} I_b R'$$

and

$$I_b = \frac{V_i}{h_{ie}}$$

with

$$V_o = -h_{fe}\frac{V_i}{h_{ie}} R'$$

so that

$$\boxed{A_v = \frac{V_o}{V_i} = -\frac{h_{fe}(R_C \| 1/h_{oe})}{h_{ie}}} \tag{8.85}$$

A_i: Assuming that $R_B \gg h_{ie}$ and $1/h_{oe} \geq 10R_C$, then $I_b \cong I_i$ and $I_o = I_c = h_{fe}I_b = h_{fe}\, I_i$ with

$$\boxed{A_i = \frac{I_o}{I_i} \cong h_{fe}} \tag{8.86}$$

EXAMPLE 8.11

For the network of Fig. 8.39, determine:

(a) Z_i.
(b) Z_o.
(c) A_v.
(d) A_i.

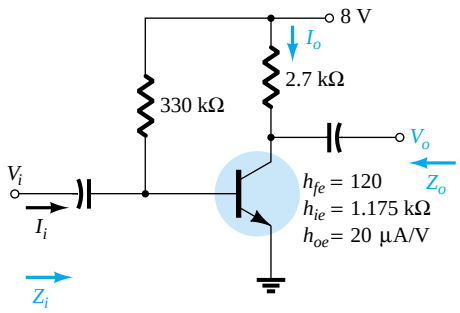


Figure 8.39 Example 8.11.

Solution

(a) $Z_i = R_B \| h_{ie} = 330\ \text{k}\Omega \| 1.175\ \text{k}\Omega$

$\cong h_{ie} = \mathbf{1.171\ k\Omega}$

(b) $r_o = \dfrac{1}{h_{oe}} = \dfrac{1}{20\ \mu\text{A/V}} = 50\ \text{k}\Omega$

$Z_o = \dfrac{1}{h_{oe}} \| R_C = 50\ \text{k}\Omega \| 2.7\ \text{k}\Omega = \mathbf{2.56\ k\Omega} \cong R_C$

(c) $Av = -\dfrac{h_{fe}(R_C \| 1/h_{oe})}{h_{ie}} = -\dfrac{(120)(2.7\ \text{k}\Omega \| 50\ \text{k}\Omega)}{1.171\ \text{k}\Omega} = \mathbf{-262.34}$

(d) $A_i \cong h_{fe} = 120$

Voltage-Divider Configuration

For the voltage-divider bias configuration of Fig. 8.40, the resulting small-signal ac equivalent network will have the same appearance as Fig. 8.38, with R_B replaced by $R' = R_1 \| R_2$.

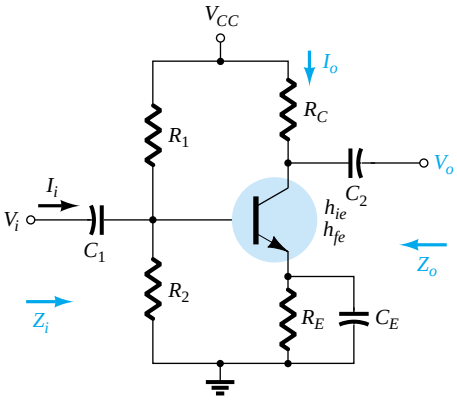


Figure 8.40 Voltage-divider bias configuration.

Z_i: From Fig. 8.38 with $R_B = R'$,

$$Z_i = R' \| h_{ie} \tag{8.87}$$

Z_o: From Fig. 8.38,

$$Z_o \cong R_C \tag{8.88}$$

A_v:

$$A_v = -\frac{h_{fe}(R_C \| 1/h_{oe})}{h_{ie}} \tag{8.89}$$

A_i:

$$A_i = -\frac{h_{fe}R'}{R' + h_{ie}} \tag{8.90}$$

Unbypassed Emitter-Bias Configuration

For the CE unbypassed emitter-bias configuration of Fig. 8.41, the small-signal ac model will be the same as Fig. 8.11, with βr_e replaced by h_{ie} and βI_b by $h_{fe}I_b$. The analysis will proceed in the same manner.

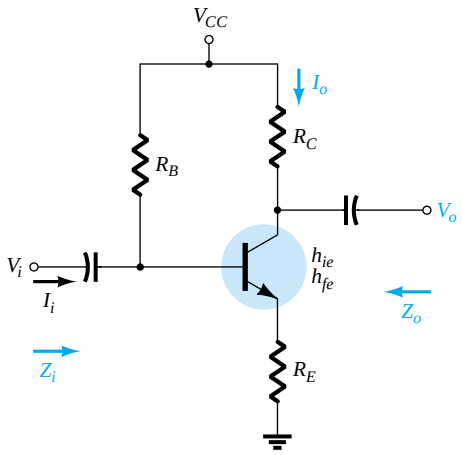


Figure 8.41 CE unbypassed emitter-bias configuration.

Z_i:

$$Z_b \cong h_{fe}\, R_E \tag{8.91}$$

and

$$Z_i = R_B \| Z_b \tag{8.92}$$

Z_o:

$$Z_o = R_C \tag{8.93}$$

A_v:

$$A_v = -\frac{h_{fe}\, R_C}{Z_b} \cong -\frac{h_{fe}R_C}{h_{fe}\, R_E}$$

and

$$A_v \cong -\frac{R_C}{R_E} \tag{8.94}$$

A_i:

$$A_i = \frac{h_{fe}R_B}{R_B + Z_b} \tag{8.95}$$

or

$$A_i = -A_v \frac{Z_i}{R_C} \tag{8.96}$$

Emitter-Follower Configuration

For the emitter-follower of Fig. 8.42, the small-signal ac model will match Fig. 8.18, with $\beta r_e = h_{ie}$ and $\beta = h_{fe}$. The resulting equations will therefore be quite similar.

Z_i:

$$Z_b \cong h_{fe} R_E \tag{8.97}$$

$$Z_i = R_B \| Z_b \tag{8.98}$$

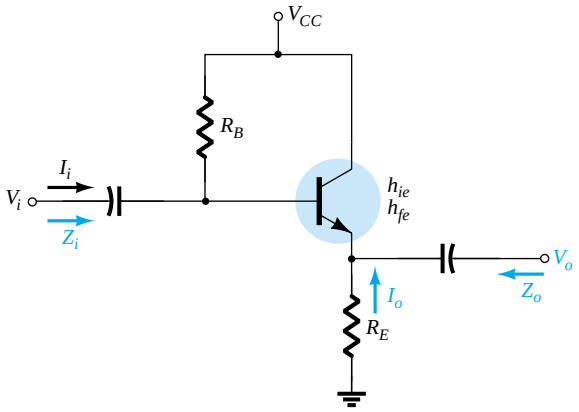


Figure 8.42 Emitter-follower configuration.

Z_o: For Z_o, the output network defined by the resulting equations will appear as shown in Fig. 8.43. Review the development of the equations in Section 8.5 and

$$Z_o = R_E \| \frac{h_{ie}}{1 + h_{fe}}$$

or since $1 + h_{fe} \cong h_{fe}$,

$$Z_o \cong R_E \| \frac{h_{ie}}{h_{fe}} \tag{8.99}$$

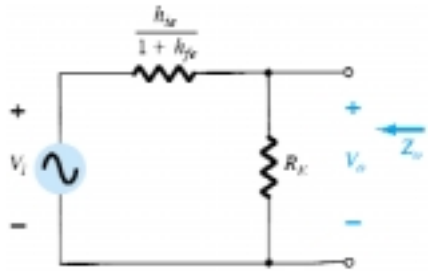


Figure 8.43 Defining Z_o for the emitter-follower configuration.

A_v: For the voltage gain, the voltage-divider rule can be applied to Fig. 8.43 as follows:

$$V_o = \frac{R_E (V_i)}{R_E + h_{ie}/(1 + h_{fe})}$$

but since $1 + h_{fe} \cong h_{fe}$,

$$A_v = \frac{V_o}{V_i} \cong \frac{R_E}{R_E + h_{ie}/h_{fe}} \tag{8.100}$$

A_i:

$$A_i = \frac{h_{fe}\, R_B}{R_B + Z_b} \tag{8.101}$$

or

$$A_i = -A_v \frac{Z_i}{R_E} \tag{8.102}$$

Common-Base Configuration

The last configuration to be examined with the approximate hybrid equivalent circuit will be the common-base amplifier of Fig. 8.44. Substituting the approximate common-base hybrid equivalent model will result in the network of Fig. 8.45, which is very similar to Fig. 8.24. From Fig. 8.45,

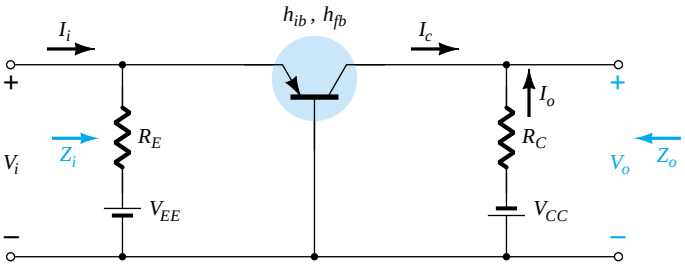


Figure 8.44 Common-base configuration.

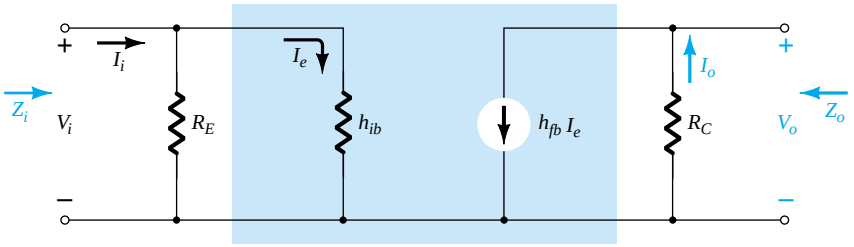


Figure 8.45 Substituting the approximate hybrid equivalent circuit into the ac equivalent network of Fig. 8.44.

Z_i:

$$Z_i = R_E \| h_{ib} \tag{8.103}$$

Z_o:

$$Z_o = R_C \tag{8.104}$$

A_v:

$$V_o = -I_o R_C = -(h_{fb} I_e) R_C$$

with $I_e = \dfrac{V_i}{h_{ib}}$ and $V_o = -h_{fb}\dfrac{V_i}{h_{ib}} R_C$

so that

$$A_v = \frac{V_o}{V_i} = -\frac{h_{fb} R_C}{h_{ib}} \tag{8.105}$$

A_i:

$$A_i = \frac{I_o}{I_i} = h_{fb} \cong -1 \tag{8.106}$$

EXAMPLE 8.12

For the network of Fig. 8.46, determine:
(a) Z_i.
(b) Z_o.
(c) A_v.
(d) A_i.

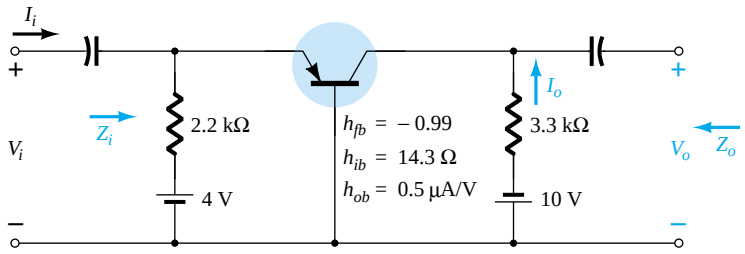


Figure 8.46 Example 8.12.

Solution

(a) $Z_i = R_E \| h_{ib} = 2.2\ \text{k}\Omega \| 14.3\ \Omega = \mathbf{14.21\ \Omega} \cong h_{ib}$

(b) $r_o = \dfrac{1}{h_{ob}} = \dfrac{1}{0.5\ \mu\text{A/V}} = 2\ \text{M}\Omega$

$$Z_o = \frac{1}{h_{ob}} \| R_C \cong R_C = \mathbf{3.3\ k\Omega}$$

(c) $A_v = -\dfrac{h_{fb} R_C}{h_{ib}} = -\dfrac{(-0.99)(3.3\ \text{k}\Omega)}{14.21} = \mathbf{229.91}$

(d) $A_i \cong h_{fb} = \mathbf{-1}$

The remaining configurations of Sections 8.1 through 8.8 that were not analyzed in this section are left as an exercise in the problem section of this chapter. It is assumed that the analysis above clearly reveals the similarities in approach using the r_e or approximate hybrid equivalent models, thereby removing any real difficulty with analyzing the remaining networks of the earlier sections.

8.10 COMPLETE HYBRID EQUIVALENT MODEL

The analysis of Section 8.9 was limited to the approximate hybrid equivalent circuit with some discussion about the output impedance. In this section, we employ the complete equivalent circuit to show the impact of h_r and define in more specific terms the

impact of h_o. It is important to realize that since the hybrid equivalent model has the same appearance for the common-base, common-emitter, and common-collector configurations, the equations developed in this section can be applied to each configuration. It is only necessary to insert the parameters defined for each configuration. That is, for a common-base configuration, h_{fb}, h_{ib}, and so on, are employed, while for a common-emitter configuration, h_{fe}, h_{ie}, and so on, are utilized. Recall that Appendix A permits a conversion from one set to the other if one set is provided and the other is required.

Consider the general configuration of Fig. 8.47 with the two-port parameters of particular interest. The complete hybrid equivalent model is then substituted in Fig. 8.48 using parameters that do not specify the type of configuration. In other words, the solutions will be in terms of h_i, h_r, h_f, and h_o. Unlike the analysis of previous sections of this chapter, the current gain A_i will be determined first since the equations developed will prove useful in the determination of the other parameters.

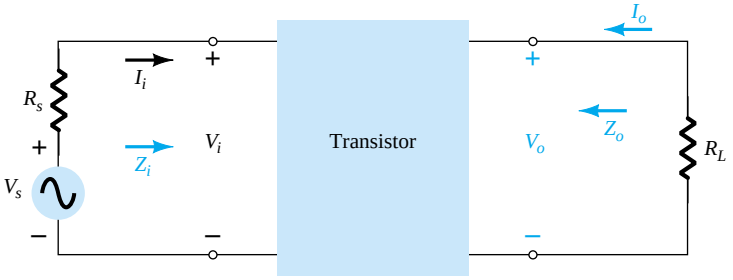


Figure 8.47 Two-port system.

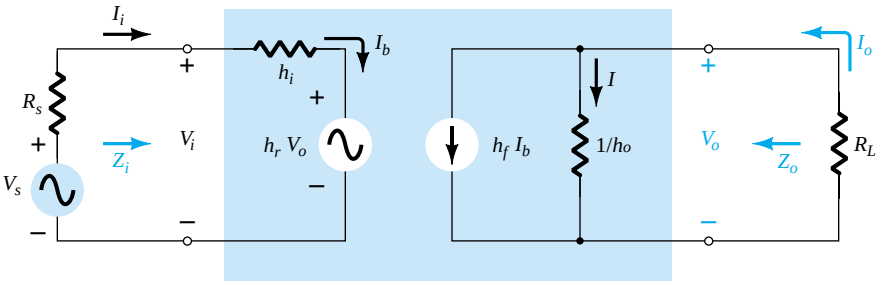


Figure 8.48 Substituting the complete hybrid equivalent circuit into the two-port system of Fig. 8.47.

Current Gain, $A_i = I_o/I_i$

Applying Kirchhoff's current law to the output circuit yields

$$I_o = h_f I_b + I = h_f I_i + \frac{V_o}{1/h_o} = h_f I_i + h_o V_o$$

Substituting $V_o = -I_o R_L$ gives us

$$I_o = h_f I_i - h_o R_L I_o$$

Rewriting the equation above, we have

$$I_o + h_o R_L I_o = h_f I_i$$

and

$$I_o(1 + h_o R_L) = h_f I_i$$

so that
$$A_i = \frac{I_o}{I_i} = \frac{h_f}{1 + h_o R_L} \tag{8.107}$$

Note that the current gain will reduce to the familiar result of $A_i = h_f$ if the factor $h_o R_L$ is sufficiently small compared to 1.

Voltage Gain, $A_v = V_o/V_i$

Applying Kirchhoff's voltage law to the input circuit results in

$$V_i = I_i h_i + h_r V_o$$

Substituting $I_i = (1 + h_o R_L) I_o / h_f$ from Eq. (8.107) and $I_o = -V_o/R_L$ from above results in

$$V_i = \frac{-(1 + h_o R_L) h_i}{h_f R_L} V_o + h_r V_o$$

Solving for the ratio V_o/V_i yields

$$A_v = \frac{V_o}{V_i} = \frac{-h_f R_L}{h_i + (h_i h_o - h_f h_r) R_L} \tag{8.108}$$

In this case, the familiar form of $A_v = -h_f R_L / h_i$ will return if the factor $(h_i h_o - h_f h_r) R_L$ is sufficiently small compared to h_i.

Input Impedance, $Z_i = V_i/I_i$

For the input circuit,

$$V_i = h_i I_i + h_r V_o$$

Substituting $V_o = -I_o R_L$

we have $V_i = h_i I_i - h_r R_L I_o$

Since $A_i = \dfrac{I_o}{I_i}$

$$I_o = A_i I_i$$

so that the equation above becomes

$$V_i = h_i I_i - h_r R_L A_i I_i$$

Solving for the ratio V_i/I_i, we obtain

$$Z_i = \frac{V_i}{I_i} = h_i - h_r R_L A_i$$

and substituting

$$A_i = \frac{h_f}{1 + h_o R_L}$$

yields
$$Z_i = \frac{V_i}{I_i} = h_i \frac{-h_f h_r R_L}{1 + h_o R_L} \tag{8.109}$$

The familiar form of $Z_i = h_i$ will be obtained if the second factor is sufficiently smaller than the first.

Output Impedance, $Z_o = V_o/I_o$

The output impedance of an amplifier is defined to be the ratio of the output voltage to the output current with the signal V_s set to zero. For the input circuit with $V_s = 0$,

$$I_i = \frac{-h_r V_o}{R_s + h_i}$$

Substituting this relationship into the following equation obtained from the output circuit yields

$$I_o = h_f I_i + h_o V_o$$

$$= \frac{-h_f h_r V_o}{R_s + h_i} + h_o V_o$$

and

$$\boxed{Z_o = \frac{V_o}{I_o} = \frac{1}{h_o - [h_f h_r/(h_i + R_s)]}} \tag{8.110}$$

In this case, the output impedance will reduce to the familiar form $Z_o = 1/h_o$ for the transistor when the second factor in the denominator is sufficiently smaller than the first.

EXAMPLE 8.13

For the network of Fig. 8.49, determine the following parameters using the complete hybrid equivalent model and compare to the results obtained using the approximate model.

(a) Z_i and Z_i'.
(b) A_v.
(c) $A_i = I_o/I_i$ and $A_i' = I_o/I_i'$.
(d) Z_o (within R_C) and Z_o' (including R_C).

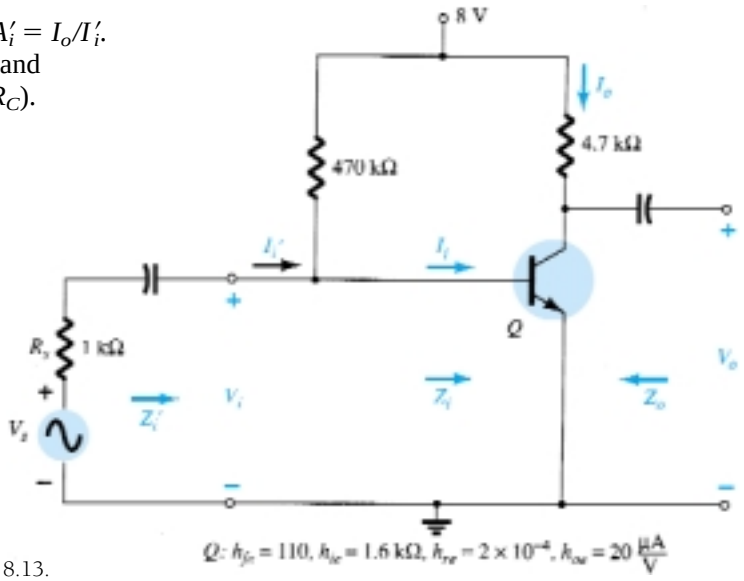


Figure 8.49 Example 8.13.

Solution

Now that the basic equations for each quantity have been derived, the order in which they are calculated is arbitrary. However, the input impedance is often a useful quantity to know and therefore will be calculated first. The complete common-emitter hybrid equivalent circuit has been substituted and the network redrawn as shown in Fig. 8.50. A Thévenin equivalent circuit for the input section of Fig. 8.50 will result in the input equivalent of Fig. 8.51 since $E_{\text{Th}} \cong V_s$ and $R_{\text{Th}} \cong R_s = 1\ \text{k}\Omega$ (a result of $R_B = 470\ \text{k}\Omega$ being much greater than $R_s = 1\ \text{k}\Omega$). In this example, $R_L = R_C$ and I_o is de-

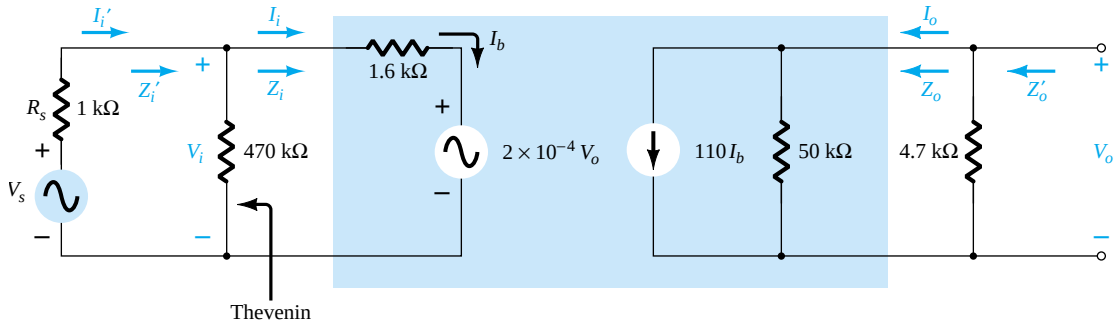


Figure 8.50 Substituting the complete hybrid equivalent circuit into the ac equivalent network of Fig. 8.49.

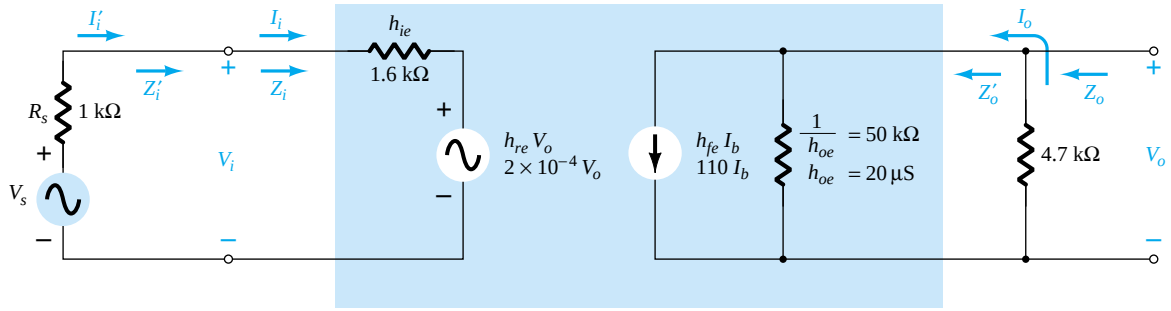


Figure 8.51 Replacing the input section of Fig. 8.50 with a Thévenin equivalent circuit.

fined as the current through R_C as in previous examples of this chapter. The output impedance Z_o as defined by Eq. (8.110) is for the output transistor terminals only. It does not include the effects of R_C. Z'_o is simply the parallel combination of Z_o and R_L. The resulting configuration of Fig. 8.51 is then an exact duplicate of the defining network of Fig. 8.48, and the equations derived above can be applied.

(a) Eq. (8.109): $$Z_i = \frac{V_i}{I_i} = h_{ie} - \frac{h_{fe}h_{re}R_L}{1 + h_{oe}R_L}$$

$$= 1.6\ \text{k}\Omega - \frac{(110)(2 \times 10^{-4})(4.7\ \text{k}\Omega)}{1 + (20\ \mu\text{S})(4.7\ \text{k}\Omega)}$$

$$= 1.6\ \text{k}\Omega - 94.52\ \Omega$$

$$= \mathbf{1.51\ k\Omega}$$

versus 1.6 kΩ using simply h_{ie}.

$$Z'_i = 470\ \text{k}\Omega \| Z_i \cong Z_i = \mathbf{1.51\ k\Omega}$$

(b) Eq. (8.108): $$A_v = \frac{V_o}{V_i} = \frac{-h_{fe}R_L}{h_{ie} + (h_{ie}h_{oe} - h_{fe}h_{re})R_L}$$

$$= \frac{-(110)(4.7\ \text{k}\Omega)}{1.6\ \text{k}\Omega + [(1.6\ \text{k}\Omega)(20\ \mu\text{S}) - (110)(2 \times 10^{-4})]4.7\ \text{k}\Omega}$$

$$= \frac{-517 \times 10^3\ \Omega}{1.6\ \text{k}\Omega + (0.032 - 0.022)4.7\ \text{k}\Omega}$$

$$= \frac{-517 \times 10^3\ \Omega}{1.6\ \text{k}\Omega + 47\ \Omega}$$

$$= \mathbf{-313.9}$$

versus -323.125 using $A_v \cong -h_{fe}R_L/h_{ie}$.

(c) Eq. (8.107): $$A_i = \frac{I_o}{I_i} = \frac{h_{fe}}{1 + h_{oe}R_L} = \frac{110}{1 + (20\ \mu\text{S})(4.7\ \text{k}\Omega)}$$

$$= \frac{110}{1 + 0.094} = \mathbf{100.55}$$

versus 110 using simply h_{fe}. Since $470\ \text{k}\Omega \gg Z_i$, $I'_i \cong I_i$ and $A'_i \cong \mathbf{100.55}$ also.

(d) Eq. (8.110): $$Z_o = \frac{V_o}{I_o} = \frac{1}{h_{oe} - [h_{fe}h_{re}/(h_{ie} + R_s)]}$$

$$= \frac{1}{20\ \mu\text{S} - [(110)(2 \times 10^{-4})/(1.6\ \text{k}\Omega + 1\ \text{k}\Omega)]}$$

$$= \frac{1}{20\ \mu\text{S} - 8.46\ \mu\text{S}}$$

$$= \frac{1}{11.54\ \mu\text{S}}$$

$$= \mathbf{86.66\ k\Omega}$$

which is greater than the value determined from $1/h_{oe} = 50\ \text{k}\Omega$.

$$Z'_o = R_C \| Z_o = 4.7\ \text{k}\Omega \| 86.66\ \text{k}\Omega = \mathbf{4.46\ k\Omega}$$

versus $4.7\ \text{k}\Omega$ using only R_C.

Note from the results above that the approximate solutions for A_v and Z_i were very close to those calculated with the complete equivalent model. In fact, even A_i was off by less than 10%. The higher value of Z_o only contributed to our earlier conclusion that Z_o is often so high that it can be ignored compared to the applied load. However, keep in mind that when there is a need to determine the impact of h_{re} and h_{oe}, the complete hybrid equivalent model must be used, as described above.

The specification sheet for a particular transistor typically provides the common-emitter parameters as noted in Fig. 7.28. The next example will employ the same transistor parameters appearing in Fig. 8.49 in a *pnp* common-base configuration to introduce the parameter conversion procedure and emphasize the fact that the hybrid equivalent model maintains the same layout.

EXAMPLE 8.14

For the common-base amplifier of Fig. 8.52, determine the following parameters using the complete hybrid equivalent model and compare the results to those obtained using the approximate model.

(a) Z_i and Z'_i.

(b) A_i and A'_i.

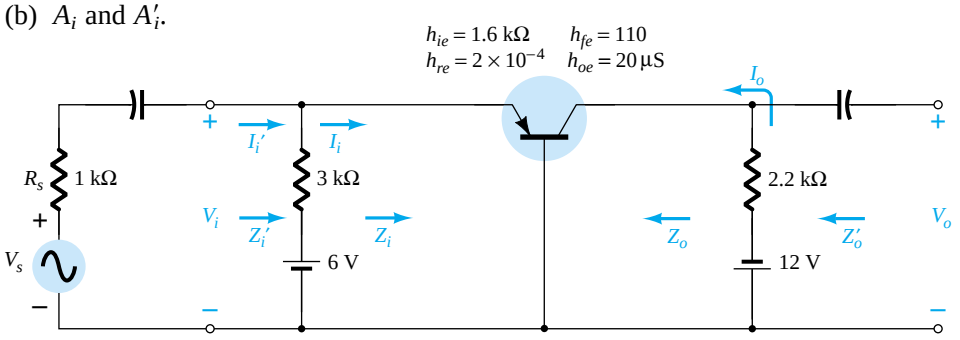


Figure 8.52 Example 8.14.

(c) A_v.
(d) Z_o and Z'_o.

Solution

The common-base hybrid parameters are derived from the common-emitter parameters using the approximate equations of Appendix A:

$$h_{ib} \cong \frac{h_{ie}}{1 + h_{fe}} = \frac{1.6\ \text{k}\Omega}{1 + 110} = \mathbf{14.41\ \Omega}$$

Note how closely the magnitude compares with the value determined from

$$h_{ib} = r_e = \frac{h_{ie}}{\beta} = \frac{1.6\ \text{k}\Omega}{110} = 14.55\ \Omega$$

$$h_{rb} \cong \frac{h_{ie}h_{oe}}{1 + h_{fe}} - h_{re} = \frac{(1.6\ \text{k}\Omega)(20\ \mu\text{S})}{1 + 110} - 2 \times 10^{-4}$$

$$= \mathbf{0.883 \times 10^{-4}}$$

$$h_{fb} \cong \frac{-h_{fe}}{1 + h_{fe}} = \frac{-110}{1 + 110} = \mathbf{-0.991}$$

$$h_{ob} \cong \frac{h_{oe}}{1 + h_{fe}} = \frac{20\ \mu\text{S}}{1 + 110} = \mathbf{0.18\ \mu S}$$

Substituting the common-base hybrid equivalent circuit into the network of Fig.

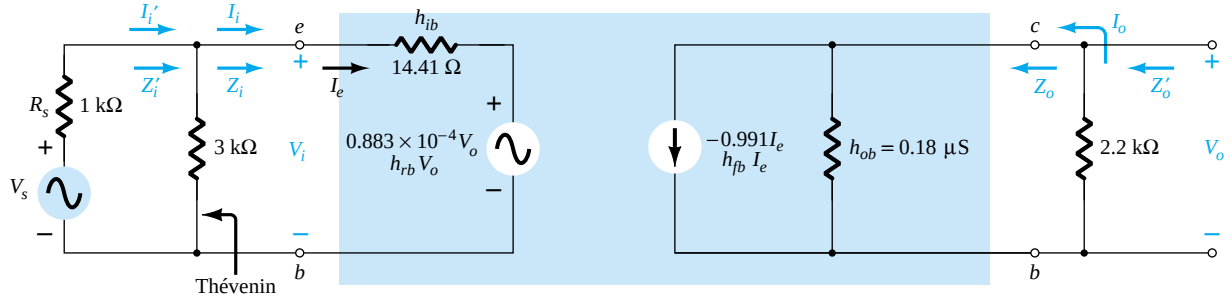


Figure 8.53 Small-signal equivalent for the network of Fig. 8.52.

8.52 will then result in the small-signal equivalent network of Fig. 8.53. The Thévenin network for the input circuit will result in $R_{\text{Th}} = 3\ \text{k}\Omega \| 1\ \text{k}\Omega = 0.75\ \text{k}\Omega$ for R_s in the equation for Z_o.

(a) Eq. (8.109): $$Z_i = \frac{V_i}{I_i} = h_{ib} - \frac{h_{fb}h_{rb}R_L}{1 + h_{ob}R_L}$$

$$= 14.41\ \Omega - \frac{(-0.991)(0.883 \times 10^{-4})(2.2\ \text{k}\Omega)}{1 + (0.18\ \mu\text{S})(2.2\ \text{k}\Omega)}$$

$$= 14.41\ \Omega + 0.19\ \Omega$$

$$= \mathbf{14.60\ \Omega}$$

versus 14.41 Ω using $Z_i \cong h_{ib}$.

$$Z'_i = 3\ \text{k}\Omega \| Z_i \cong Z_i = \mathbf{14.60\ \Omega}$$

(b) Eq. (8.107): $$A_i = \frac{I_o}{I_i} = \frac{h_{fb}}{1 + h_{ob}R_L}$$

$$= \frac{-0.991}{1 + (0.18\ \mu\text{S})(2.2\ \text{k}\Omega)}$$

$$= \mathbf{-0.991} = h_{fb}$$

Since 3 kΩ $\gg Z_i$, $I'_i \cong I_i$ and $A'_i = I_o/I'_i \cong -1$ also.

(c) Eq. (8.108): $$A_v = \frac{V_o}{V_i} = \frac{-h_{fb}R_L}{h_{ib} + (h_{ib}h_{ob} - h_{fb}h_{rb})R_L}$$

$$= \frac{-(-0.991)(2.2\ \text{k}\Omega)}{14.41\ \Omega + [(14.41\ \Omega)(0.18\ \mu\text{S}) - (-0.991)(0.883 \times 10^{-4})]2.2\ \text{k}\Omega}$$

$$= \mathbf{149.25}$$

versus 151.3 using $A_v \cong -h_{fb}R_L/h_{ib}$.

(d) Eq. (8.110): $$Z_o = \frac{1}{h_{ob} - [h_{fb}h_{rb}/(h_{ib} + R_s)]}$$

$$= \frac{1}{0.18\ \mu\text{S} - [(-0.991)(0.883 \times 10^{-4})/(14.41\ \Omega + 0.75\ \text{k}\Omega)]}$$

$$= \frac{1}{0.295\ \mu\text{S}}$$

$$= \mathbf{3.39\ M\Omega}$$

versus 5.56 MΩ using $Z_o \cong 1/h_{ob}$. For Z'_o as defined by Fig. 8.53:

$$Z'_o = R_C \| Z_o = 2.2\ \text{k}\Omega \| 3.39\ \text{M}\Omega = \mathbf{2.199\ k\Omega}$$

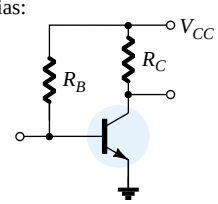
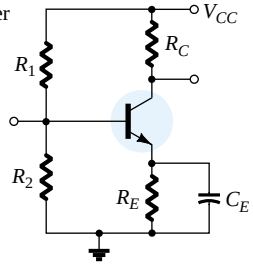
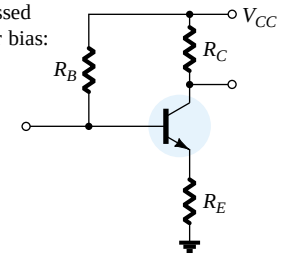
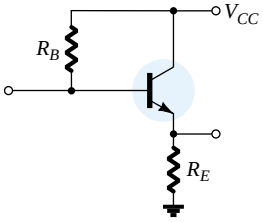
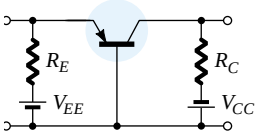
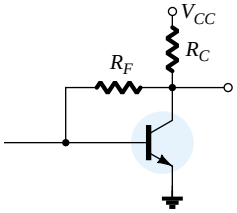
versus 2.2 kΩ using $Z'_o \cong R_C$.

8.11 SUMMARY TABLE

Now that the most familiar configurations of the small-signal transistor amplifiers have been introduced, Table 8.1 is presented to review the general characteristics of each for immediate recall. It must be absolutely clear that the values listed are simply typical values to establish a basis for comparison. The levels obtained in an actual analysis will most likely be different, and certainly different from one configuration to another. Being able to repeat most of the information in the table is an important first step in developing a general familiarity with the subject matter. For instance, one should now be able to state with some assurance that the emitter-follower configuration typically has a high input impedance, low output impedance, and a voltage gain slightly less than 1. There should be no need to perform a variety of calculations to recall salient facts such as those above. For the future, it will permit the study of a network or system without becoming mathematically involved. The function of each component of a design will become increasingly familiar as general facts such as those above become part of your background.

One obvious advantage of being able to recall general facts like the above is an ability to check the results of a mathematical analysis. If the input impedance of a common-base configuration is in the kilohm range, there is good reason to recheck the analysis. However, on the other side of the coin, a result of 22 Ω suggests that the analysis may be correct.

TABLE 8.1 Relative Levels for the Important Parameters of the CE, CB, and CC Transistor Amplifiers

Configuration	Z_i	Z_o	A_v	A_i
Fixed-bias: 	Medium (1 kΩ) $= R_B \| \beta r_e$ $\cong \beta r_e$ $(R_B \geq 10\beta r_e)$	Medium (2 kΩ) $= R_C \| r_o$ $\cong R_C$ $(r_o \geq 10R_C)$	High (−200) $= -\frac{(R_C \| r_o)}{r_e}$ $\cong -\frac{R_C}{r_e}$ $(r_o \geq 10R_C)$	High (100) $= \frac{\beta R_B r_o}{(r_o + R_C)(R_B + \beta r_e)}$ $\cong \beta$ $(r_o \geq 10R_C,$ $R_B \geq 10\beta r_e)$
Voltage-divider bias: 	Medium (1 kΩ) $= R_1 \| R_2 \| \beta r_e$	Medium (2 kΩ) $= R_C \| r_o$ $\cong R_C$ $(r_o \geq 10R_C)$	High (−200) $= -\frac{R_C \| r_o}{r_e}$ $\cong -\frac{R_C}{r_e}$ $(r_o \geq 10R_C)$	High (50) $= \frac{\beta(R_1 \| R_2) r_o}{(r_o + R_C)(R_1 \| R_2 + \beta r_e)}$ $\cong \frac{\beta \| (R_1 \| R_2)}{R_1 \| R_2 + \beta r_e}$ $(r_o \geq 10R_C)$
Unbypassed emitter bias: 	High (100 kΩ) $= R_B \| Z_b$ $Z_b \cong \beta(r_e + R_E)$ $\cong R_B \| \beta R_E$ $(R_E \gg r_e)$	Medium (2 kΩ) $= R_C$ (any level of r_o)	Low (−5) $= -\frac{R_C}{r_e + R_E}$ $\cong -\frac{R_C}{R_E}$ $(R_E \gg r_e)$	High (50) $\cong -\frac{\beta R_B}{R_B + Z_b}$
Emitter-follower: 	High (100 kΩ) $= R_B \| Z_b$ $Z_b \cong \beta(r_e + R_E)$ $\cong R_B \| \beta R_E$ $(R_E \gg r_e)$	Low (20 Ω) $= R_E \| r_e$ $\cong r_e$ $(R_E \gg r_e)$	Low ($\cong 1$) $= \frac{R_E}{R_E + r_e}$ $\cong 1$	High (−50) $\cong -\frac{\beta R_B}{R_B + Z_b}$
Common-base: 	Low (20 Ω) $= R_E \| r_e$ $\cong r_e$ $(R_E \gg r_e)$	Medium (2 kΩ) $= R_C$	High (200) $\cong \frac{R_C}{r_e}$	Low (−1) $\cong -1$
Collector feedback: 	Medium (1 kΩ) $= \frac{r_e}{\frac{1}{\beta} + \frac{R_C}{R_E}}$ $(r_o \geq 10R_C)$	Medium (2 kΩ) $\cong R_C \| R_F$ $(r_o \geq 10R_C)$	High (−200) $\cong -\frac{R_C}{r_e}$ $(r_o \geq 10R_C)$ $R_F \gg R_C)$	High (50) $= \frac{\beta R_F}{R_F + \beta R_C}$ $\cong \frac{R_F}{R_C}$

8.12 TROUBLESHOOTING

Although the terminology *troubleshooting* suggests that the procedures to be described are designed simply to isolate a malfunction, it is important to realize that the same techniques can be applied to ensure that a system is operating properly. In any case, the testing, checking, or isolating procedures require an understanding of what to expect at various points in the network in both the dc and ac domains. In most cases, a network operating correctly in the dc mode will also behave properly in the ac domain. In addition, a network providing the expected ac response is most likely biased as planned. In a typical laboratory setting, both the dc and ac supplies are applied and the ac response at various points in the network is checked with an oscilloscope as shown in Fig. 8.54. Note that the black (gnd) lead of the oscilloscope is connected directly to ground and the red lead is moved from point to point in the network, providing the patterns appearing in Fig. 8.54. The vertical channels are set in the ac mode to remove any dc component associated with the voltage at a particular point. The small ac signal applied to the base is amplified to the level appearing from collector to ground. Note the difference in vertical scales for the two voltages. There is no ac

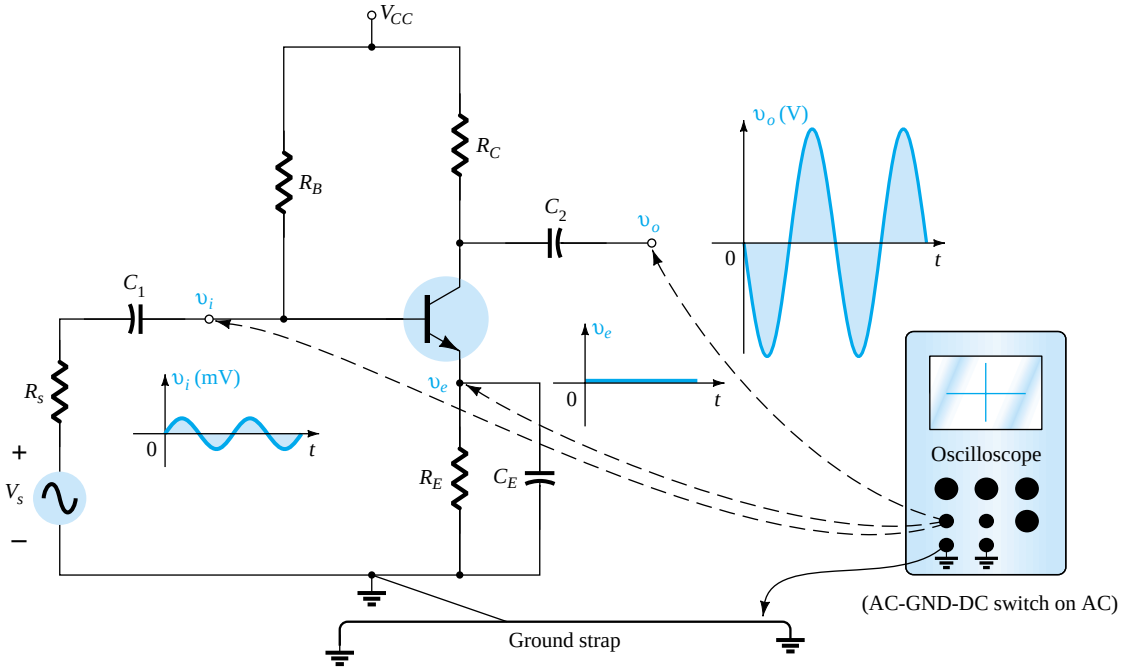


Figure 8.54 Using the oscilloscope to measure and display various voltages of a BJT amplifier.

response at the emitter terminal due to the short-circuit characteristics of the capacitor at the applied frequency. The fact that v_o is measured in volts and v_i in millivolts suggests a sizable gain for the amplifier. In general, the network appears to be operating properly. If desired, the dc mode of the multimeter could be used to check V_{BE} and the levels of V_B, V_{CE}, and V_E to review whether they lie in the expected range. Of course, the oscilloscope can also be used to compare dc levels simply by switching to the dc mode for each channel.

Needless to say, a poor ac response can be due to a variety of reasons. In fact, there may be more than one problem area in the same system. Fortunately, however, with time and experience, the probability of malfunctions in some areas can be predicted and an experienced person can isolate problem areas fairly quickly.

In general, there is nothing mysterious about the general troubleshooting process. If you decide to follow the ac response, it is good procedure to start with the applied

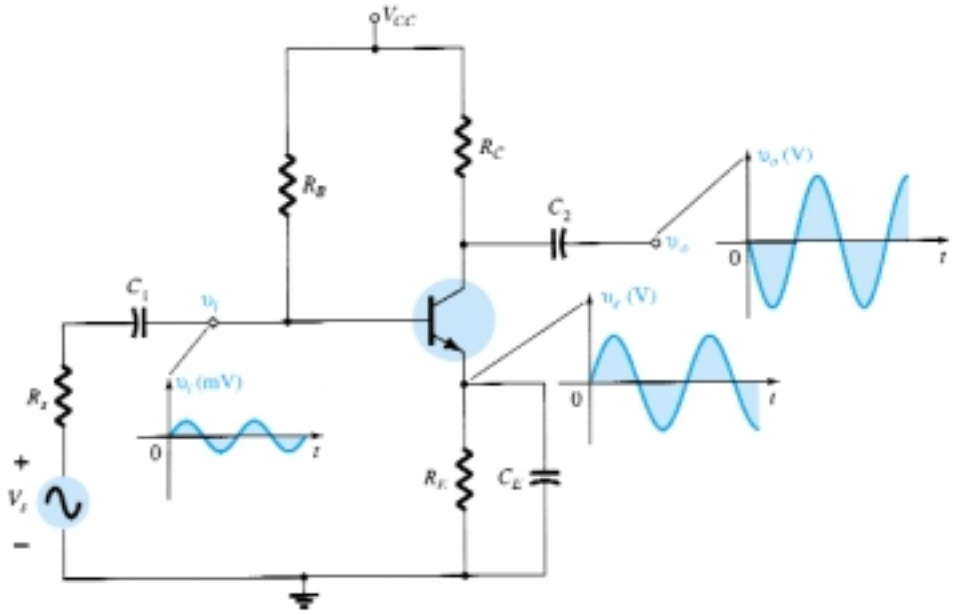


Figure 8.55 The waveforms resulting from a malfunction in the emitter area.

signal and progress through the system toward the load, checking critical points along the way. An unexpected response at some point suggests that the network is fine up to that area, thereby defining the region that must be investigated further. The waveform obtained on the oscilloscope will certainly help in defining the possible problems with the system.

If the response for the network of Fig. 8.54 is as appears in Fig. 8.55, the network has a malfunction that is probably in the emitter area. An ac response across the emitter is unexpected, and the gain of the system as revealed by v_o is much lower. Recall for this configuration that the gain is much greater if R_E is bypassed. The response obtained suggests that R_E is not bypassed by the capacitor and the terminal connections of the capacitor and the capacitor itself should be checked. In this case, a checking of the dc levels will probably not isolate the problem area since the capacitor has an "open-circuit" equivalent for dc. In general, a prior knowledge of what to expect, a familiarity with the instrumentation, and most important, experience are all factors that contribute to the development of an effective approach to the art of troubleshooting.

8.13 PSPICE WINDOWS

Voltage-Divider Configuration Using the Software Transistor Parameters

Now that the basic maneuvers for developing the network on the schematics grid have been introduced, the current description will concentrate on the variations introduced by the ac analysis.

Using schematics, the network of Fig. 8.9 (Example 8.2) is developed as shown in Fig. 8.56. Note the ac source of 1 mV and the printer symbol at the output terminal of the network.

The sinusoidal ac source is listed in the **SOURCE.slb** library as **VSIN.** Once

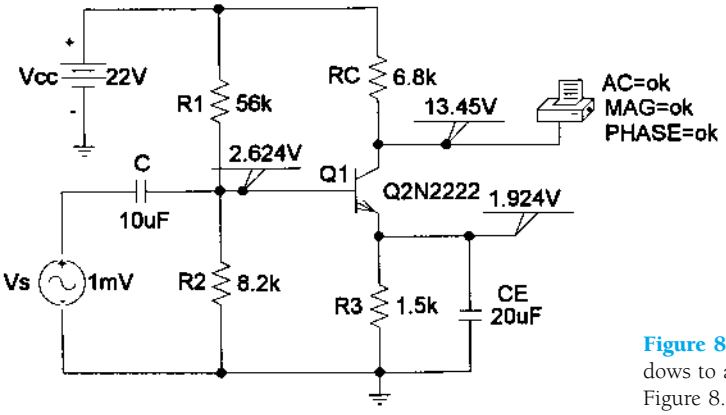


Figure 8.56 Using PSpice Windows to analyze the network of Figure 8.9 (Example 8.2).

placed on the diagram, double-clicking the symbol will result in the **PartName: VSIN** dialog box with a list of options. Each choice can be made by double-clicking the desired quantity, which will then appear in the **Name** and **Value** rectangles at the top of the box. The cursor appears in the **Value** box, and the desired value can be entered. After each entry, be sure to **Save Attr** to save the entered attribute. If done properly, the assigned value will appear in the listing.

For our analysis, the following choices will be made:

VAMPL = 1mV (the peak value of the sinusoidal signal).

FREQ = 10kHz (the frequency of interest).

PHASE = 0 (no initial phase angle for the sinusoidal signal).

VOFF = 0 (no dc offset voltage for the sinusoidal signal).

AC = 1mV.

If you want to display the value of the ac signal, simply click on **Change Display** after saving the attribute. For instance, if AC = 1mV was just saved and **Change Display** was chosen, a **Change Attribute** dialog box would appear. Since AC is the name and 1mV the value, choose **Value Only,** and only the 1mV will be displayed after the sequence **OK-OK.**

The printer symbol on the collector of the transistor is listed as **VPRINT1** under the **SPECIAL.slb** library. When placed on the schematic, it dictates that the ac voltage at that point will be printed in the output file (*.out). Double-clicking on the printer symbol will result in a **PRINT1** dialog box, within which the following choices should be made:

AC = ok.

MAG = ok.

PHASE = ok.

After each entry, be sure to **Save Attr** or the computer will remind you. The above choices can be listed next to the printer symbol on the schematic by simply clicking the **Change Display** option and choosing the **Display Value and Name** for each item.

The transistor is obtained through the sequence **Get New Part** icon-**Libraries-EVAL.slb-Q2N2222-OK-Place & Close.** Since we will want the parameters of the transistor to match those of the example as closely as possible, one must first click on the transistor to put it in the active mode (red) and then choose **Edit-Model-Edit Instance Model (Text).** Next, the beta (**Bf**) is set to 90 and **Is** is set to 2E-15A to result in a base-to-emitter voltage close to 0.7 V. This value of *I* is the result of nu-

merous runs of the network to find that value of I_S that provided a level of V_{BE} closest to 0.7 V. For the remainder of this text, however, this chosen level of I_S will remain the same. In most cases, it provides the desired results.

VIEWPOINTs have been inserted to display the three dc voltages of interest. Since it has been used recently, **VIEWPOINT** can be found in the scroll listing at the top right of the menu bar rather than by returning to the library listing.

Choosing the **Setup Analysis** icon will result in the **Analysis Setup** dialog box, in which the **AC Sweep** must be chosen because of the applied ac source. Clicking on **AC Sweep** will result in an **AC Sweep and Noise Analysis** dialog box, in which **Linear** is chosen along with **Total Pts:** 1, **Start Freq:** 10kHz, and **End Freq:** 10kHz. The result will be an analysis at only one frequency. Our initial interest will simply be in the magnitude of the quantities of interest and not their shape or appearance. Therefore, we should turn to **Analysis-Probe Setup** and choose **Do not auto-run Probe** to save time getting to the desired results.

Clicking the **Analysis** icon will result in a **PSpiceAD** dialog box that will indicate the **AC Analysis** is finished. Note also the listing of the frequency applied at the bottom of the dialog box. Within this box, if we choose **File** followed by **Examine Output,** we will obtain a lengthy listing of input and output data on the analyzed network. Specific headings are duplicated in Fig. 8.57. Under **Schematics Netlist,** the nodes assigned to the network are revealed. Note that ground is always defined as the 0 node and the assumed node of higher potential listed first. The transistor is listed in the order Collector-Base-Emitter. Under **BJT MODEL PARAMETERS,** the defining parameters of the device are listed with the set values of I_S = 2E-15A and β = 90. Under **SMALL-SIGNAL BIAS SOLUTION,** the dc levels at the various nodes are revealed, which compare directly with the **VIEWPOINT** values. In particular, note that V_{BE} is exactly 0.7 V.

The next listing, **OPERATING POINT INFORMATION,** reveals that even though beta of the **BJT MODEL PARAMETERS** listing was set at 90, the operating conditions of the network resulted in a dc beta of 48.3 and an ac beta of 55. Fortunately, however, the voltage-divider configuration is less sensitive to changes in beta in the dc mode, and the dc results are excellent. However, the drop in ac beta had an effect on the resulting level of V_o: 296.1 mV versus the hand-written solution (with r_o = 50 kΩ) of 324.3 mV—a 9% difference. The results are certainly close, but probably not as close as one would like. A closer result (within 7%) could be obtained by setting all the parameters of the device except I_S and beta to zero. However, for the moment, the impact of the remaining parameters has been demonstrated, and the results will be accepted as sufficiently close to the hand-written levels. Later in this chapter, an ac model for the transistor will be introduced with results that will be an exact match with the hand-written solution. The phase angle is −178° versus the ideal of −180°—a very close match.

A plot of the output waveform can be obtained using the **Probe** option. The sequence **Analysis-Probe Setup-Automatically run Probe after simulation-OK** will result in a **MicroSim Probe** screen when the **Analysis** icon is chosen. However, if we follow this procedure without setting the horizontal scale, we will simply end up with a plot point of 296 mV at a frequency of 10 kHz. The horizontal scale is set by the sequence **Analysis-Setup-Transient** with the **AC Sweep** disabled. Clicking the **Transient** option will result in a **Transient** dialog box, in which a number of choices have to be made based on the waveform to be viewed. The period of the applied signal of 10 kHz is 0.1 ms = 100 μs. The **Print Step** option refers to the time interval between printing or plotting the results of the transient analysis. For our example, we will choose 1 μs to provide 100 plot points per cycle. The **Final Time** is the last instant the network's response will be determined. Our choice is 500 μs or 0.5 ms to provide five full cycles of the waveform. The **No-Print Delay** was chosen as 0 since all the capacitors are essentially short cir-

```
 ****     CIRCUIT DESCRIPTION
******************************************************************************
* Schematics Netlist *

V_V1         $N_0001 0 22
R_R1         $N_0001 $N_0002  56k
R_R4         $N_0002 0  8.2k
R_R2         $N_0001 $N_0003  6.8k
R_R3         $N_0004 0  1.5k
C_C1         $N_0004 0  20u
C_C2         $N_0005 $N_0002  10u
V_V2         $N_0005 0  AC 1m
+SIN 0 1m 10k 0 0 0

.PRINT         AC
+ VM([$N_0003])
+ VP([$N_0003])
Q_Q1         $N_0003 $N_0002 $N_0004 Q2N2222-X

**** RESUMING edc8a.cir ****

 ****     BJT MODEL PARAMETERS
******************************************************************************
               Q2N2222-X
               NPN
         IS     2.000000E-15
         BF    90
         NF     1
        VAF    74.03
        IKF      .2847
        ISE    14.340000E-15
         NE     1.307
         BR     6.092
         NR     1
         RB    10
         RC     1
        CJE    22.010000E-12
        MJE      .377
        CJC     7.306000E-12
        MJC      .3416
         TF   411.100000E-12
        XTF     3
        VTF     1.7
        ITF      .6
         TR    46.910000E-09
        XTB     1.5

 ****     SMALL SIGNAL BIAS SOLUTION       TEMPERATURE =   27.000 DEG C
******************************************************************************
 NODE   VOLTAGE     NODE   VOLTAGE     NODE   VOLTAGE     NODE   VOLTAGE

($N_0001)   22.0000                  ($N_0002)    2.6239

($N_0003)   13.4530                  ($N_0004)    1.9244

($N_0005)    0.0000


    VOLTAGE SOURCE CURRENTS
    NAME         CURRENT

    V_V1        -1.603E-03
    V_V2         0.000E+00

    TOTAL POWER DISSIPATION   3.53E-02  WATTS

 ****     OPERATING POINT INFORMATION      TEMPERATURE =   27.000 DEG C
******************************************************************************
**** BIPOLAR JUNCTION TRANSISTORS


NAME         Q_Q1
MODEL        Q2N2222-X
IB           2.60E-05
IC           1.26E-03
VBE          6.99E-01
VBC         -1.08E+01
VCE          1.15E+01
BETADC       4.83E+01
GM           4.84E-02
RPI          1.14E+03
RX           1.00E+01
RO           6.75E+04
CBE          5.78E-11
CBC          2.87E-12
CJS          0.00E+00
BETAAC       5.50E+01
CBX          0.00E+00
FT           1.27E+08

 ****     AC ANALYSIS                      TEMPERATURE =   27.000 DEG C
******************************************************************************
  FREQ        VM($N_0003) VP($N_0003)

   1.000E+04   2.961E-01  -1.780E+02
```

Figure 8.57 Output file for the network of Figure 8.56.

cuits at 10 kHz. If we felt there was a transient phase between energizing the network and reaching a steady-state response, the **No-Print Delay** could be used to effectively eliminate this period of time. The last choice of **Step Ceiling** sets a maximum time period between response calculations for the system, which we will set at 1 μs. The time between calculations will be adjusted internally by the software package to ensure sufficient data at times when the response may change faster than usual. However, they will never be separated by a time period greater than that set by the **Step Ceiling.**

After **Simulation,** a **MicroSim Probe** screen will appear showing only the horizontal scale from 0 to 500 μs as specified in the **Transient** dialog box. To obtain a waveform, one can either choose **Trace** on the menu bar or the **Trace** icon (red pattern on a black axis). If the **Trace** on the menu bar is chosen, one must follow with **Add,** and the **Add Traces** dialog box will appear. Using the icon results in the dialog box immediately. Now one must choose the waveform to be displayed from the list of **Simulation Output Variables.** Since we want the collector-to-emitter voltage of the transistor, **V(Q1:c)**—an option under **Alias** names—will be chosen, followed by an **OK.** The result is the waveform of Fig. 8.58, with the waveform riding on the

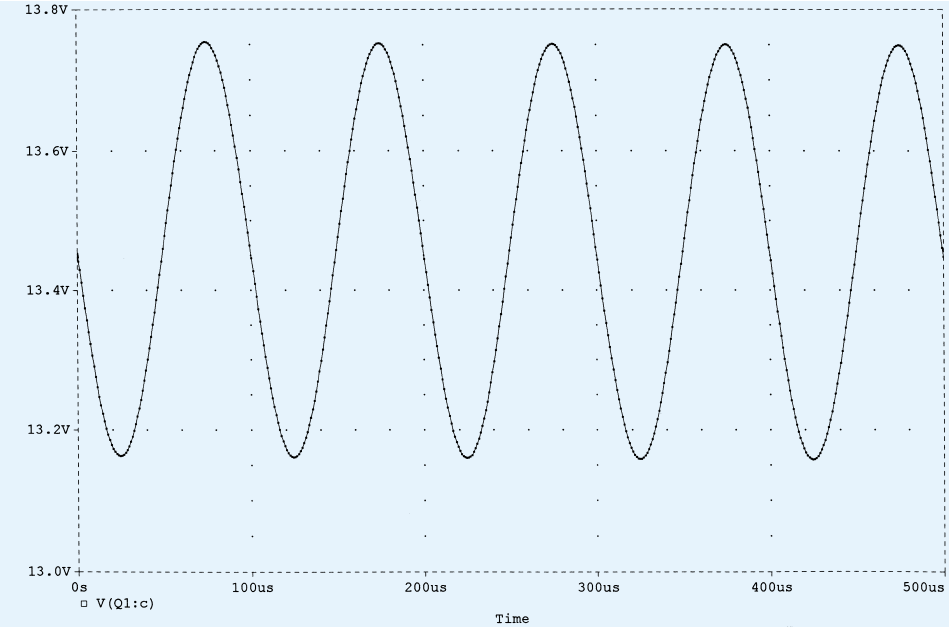


Figure 8.58 Voltage v_c for the network of Figure 8.56.

dc level of 13.45 V. The range of the vertical axis was automatically chosen by the computer. Five full cycles of the output waveform are displayed (as we expected) with 100 data points for each cycle. If you would like to see the data points (as shown in Fig. 8.58), simply turn to **Tools-Options-Probe Options** and choose **Mark Data Points.** Click **OK,** and the data points will appear. Using the scale of the graph, the peak-to-peak value of the curve is approximately 13.76 V − 13.16 V = 0.6 V = 600 mV, resulting in a peak value of 300 mV. Since a 1-mV signal was applied, the gain is 300, or very close to the values displayed above.

If a comparison is to be made between the input and output voltages on the same graph, the **Add Y-Axis** option under **Plot** can be chosen. After it is triggered, choose

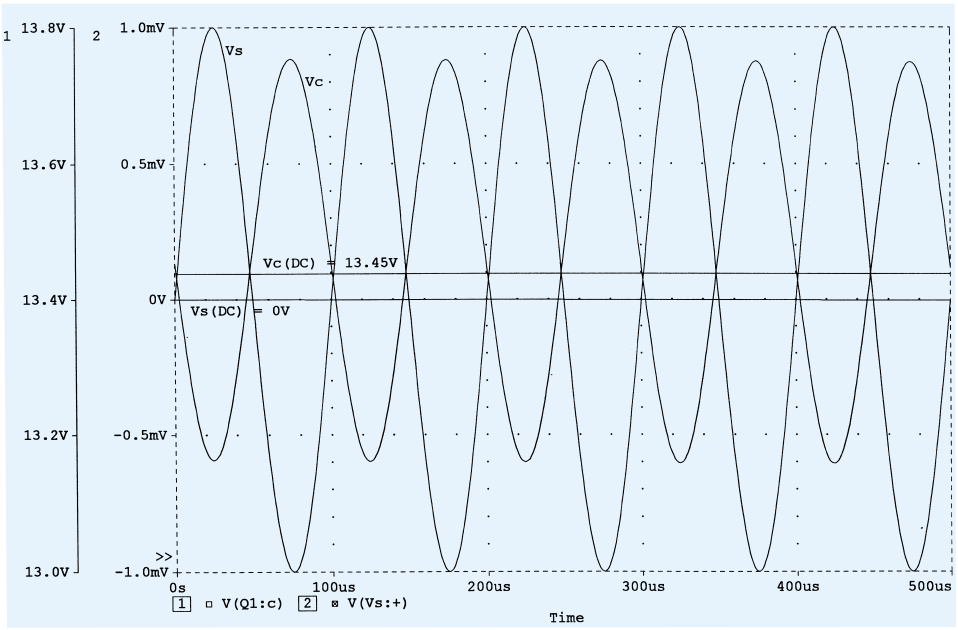


Figure 8.59 The voltages v_c and v_s for the network of Figure 8.56.

the **Add Trace** icon and select **V(Vs:+).** The result is that both waveforms will appear on the same screen, each with their own vertical scale. Labels can be added to the waveforms as shown in Fig. 8.59 using **Tools-Label-Text.** A **Text Label** dialog box will appear, in which the desired text can be entered. Click **OK,** and it can be placed with the mouse in any location on the graph. Lines can also be added with **Tools-Label-Line.** A pencil will appear, which can be used to draw the line with a left-click at the starting point and another click when the line is in place. Each plot

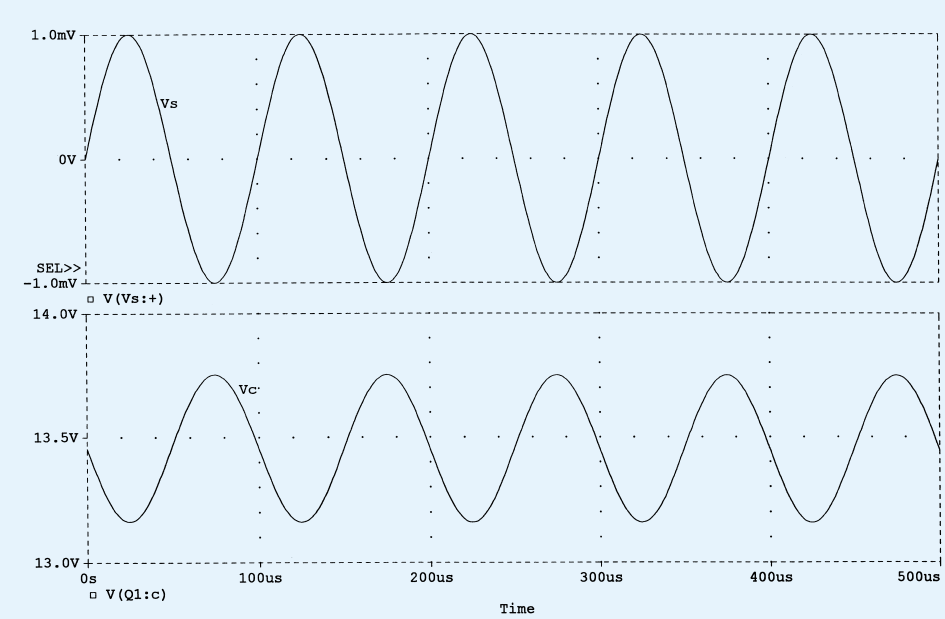


Figure 8.60 Two separate plots of v_c and v_s in Figure 8.56.

can be printed with **File-Print-Copies-OK.**

If two separate graphs are preferred, we can choose the **Plot** option and select **Add Plot** after **V** has been displayed. Upon selection, another graph will appear, waiting for the next choice. The sequence **Trace-Add-V(Vs:+)** will then result in the graphs of Fig. 8.60. The labels **Vs** and **Vc** were added using the **Tools** option. If further operations are to be performed on either graph, the **SEL**′+ defines the active plot.

The last waveform of this section will demonstrate the use of the **Cursor** option that can be called up using the **Tools** menu choice or the **Cursor Point** icon (having a graph with an arrow drawn from the graph to the vertical axis). The sequence **Tools-Cursor-Display** will result in a line at the dc level of 13.453 V, as shown in the dialog box at the bottom right of the graph of Fig. 8.61. Left-clicking on the mouse once will result in a horizontal and vertical line intersecting at some point on the curve. By clicking on the vertical line and holding it down, the vertical line and corresponding horizontal line (on the graph) can be moved across the waveform. At each point, the vertical and horizontal intersections will appear in the dialog box. If moved to the first peak value, **A1** will be at 13.754 V and 74.825 μs. By right-clicking on the mouse, a second intersection, defined by **A2,** appears, which also has its location registered in the dialog box. These intersecting lines are moved by holding down

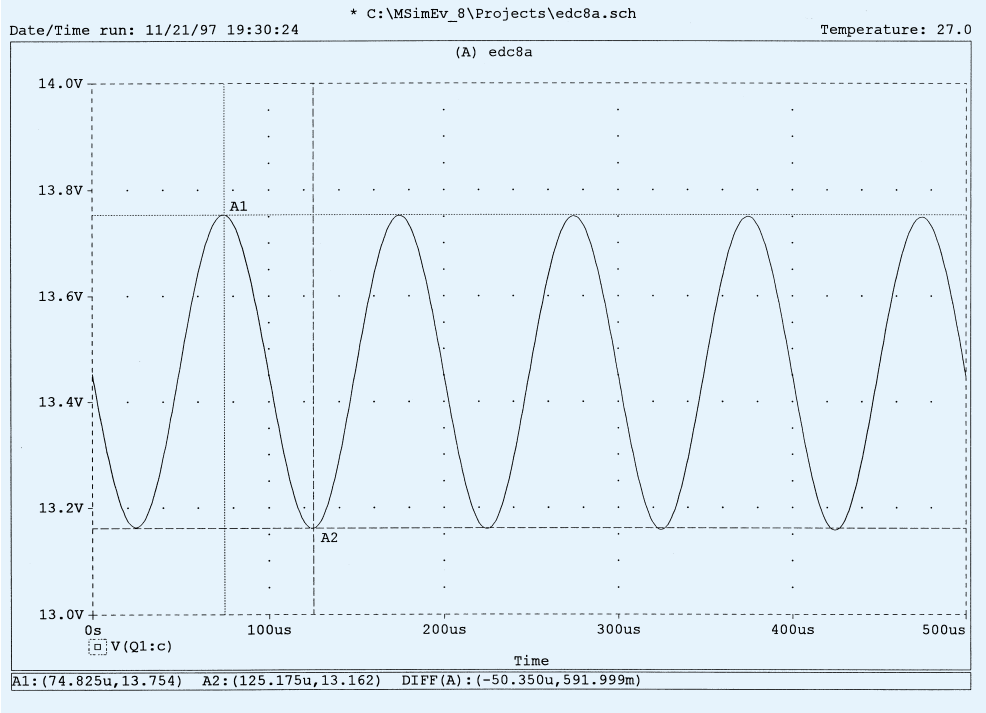


Figure 8.61 Demonstrating the use of cursors to read specific points on a plot.

the right side of the mouse. The remaining information on the third line of the box is the difference between the two intersections on the two axis. If **A2** is set at the bottom of the waveform as shown in Fig. 8.61, it will read 13.162 V at 125.17 μs, resulting in a difference between the two of 591.999 mV, or 0.592 V vertically and 50.35 μs horizontally. This is as expected, because the peak-to-peak value matches the 2×0.296 V = 0.592 V obtained earlier. The time interval is essentially 1/4 of the total period (200 μs) of the waveform. The labels **A1** and **A2** were added using the **Tools-Label-Text** sequence or the **ABC** text icon.

The peak and minimum values for the graph of Fig. 8.61 can also be found using the icons appearing in the top right region of the menu bar. Once the desired wave-

form is obtained and the sequence **Tools-Cursor-Display** applied or the **Toggle cursor** icon (the icon in the center region of the menu bar with the black dashed axis and red curve passing through the origin) is chosen, the six icons to the right of the **Toggle cursor** icon will change to a color pattern indicating they are ready for use. Clicking on the icon with the intersection at the top will automatically place the **A1** intersection at the top of the curve. Clicking the next icon to the right will place the intersection at the bottom (trough) of the curve. The next icon will place the intersection at the steepest slope and the next at the minimum value (matching the trough value).

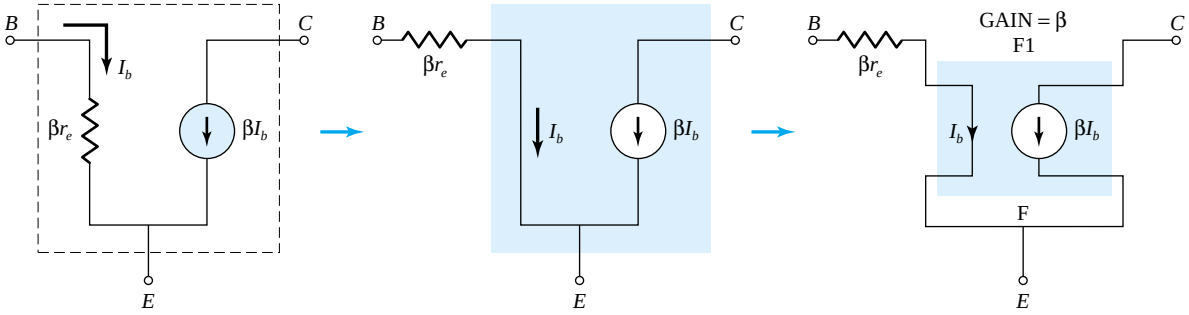


Figure 8.62 Using a controlled source to represent the transistor of Figure 8.56.

Voltage-Divider Configuration—Controlled Source Substitution

The results obtained for any analysis using the transistors provided in the software package will always be different from those obtained with an equivalent model that only includes the effect of beta and r_e. This was demonstrated for the network of Fig. 8.56. If a solution is desired that is limited to the approximate model, then the transistor must be represented by a model such as appearing in Fig. 8.62.

For Example 8.2, β is 90, with $\beta r_e = 1.66\ \text{k}\Omega$. The current controlled current source (**CCCS**) is found in the **ANALOG.slb** library as **Part F.** When you click on **F,** the **Description** above will read **Current-controlled current source.** After **OK-Place & Close,** the graphical symbol for the **CCCS** will appear on the screen as shown in Fig. 8.63. Since βr_e does not appear within the basic structure of the **CCCS,** it must be added in series with the controlling current indicated by the arrow on the

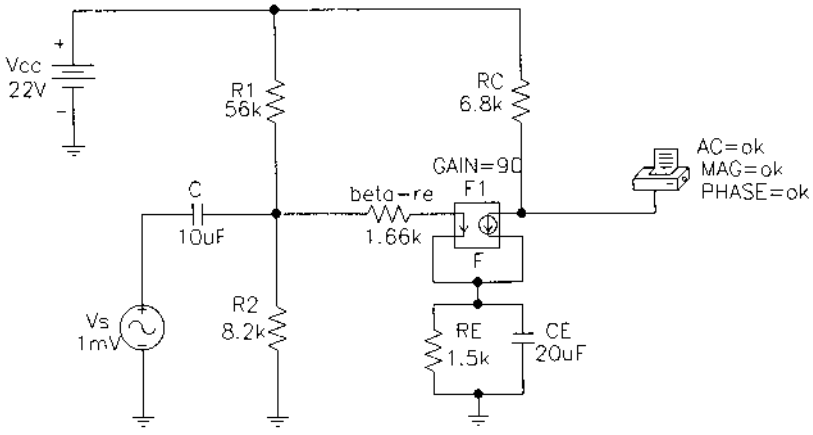


Figure 8.63 Substituting the controlled source of Figure 8.62 for the transistor of Figure 8.56.

§ 8.2 Common-Emitter Fixed-Bias Configuration

1. For the network of Fig. 8.64:
(a) Determine Z_i and Z_o.
(b) Find A_v and A_i.
(c) Repeat part (a) with $r_o = 20\ \text{k}\Omega$.
(d) Repeat part (b) with $r_o = 20\ \text{k}\Omega$.

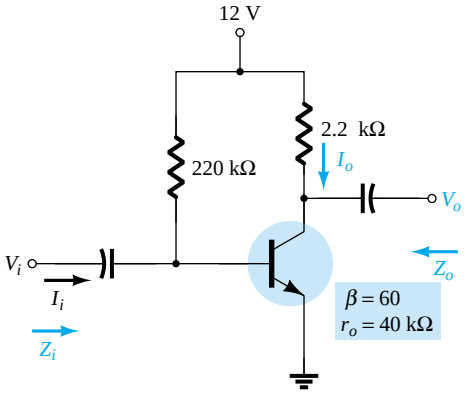


Figure 8.64 Problems 1 and 21

2. For the network of Fig. 8.65, determine V_{CC} for a voltage gain of $A_v = -200$.

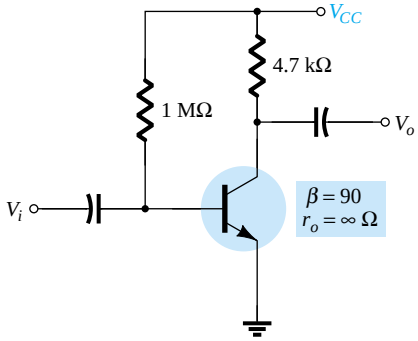


Figure 8.65 Problem 2

* **3.** For the network of Fig. 8.66:
(a) Calculate I_B, I_C, and r_e.
(b) Determine Z_i and Z_o.
(c) Calculate A_v and A_i.
(d) Determine the effect of $r_o = 30\ \text{k}\Omega$ on A_v and A_i.

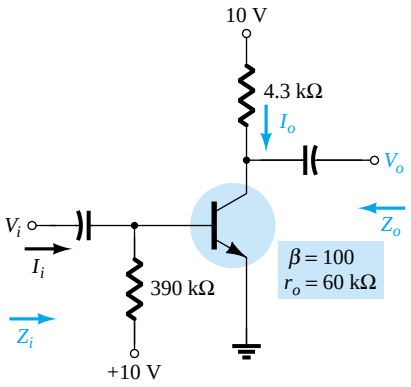


Figure 8.66 Problem 3

§ 8.3 Voltage-Divider Bias

4. For the network of Fig. 8.67:
(a) Determine r_e.
(b) Calculate Z_i and Z_o.
(c) Find A_v and A_i.
(d) Repeat parts (b) and (c) with $r_o = 25\ \text{k}\Omega$.

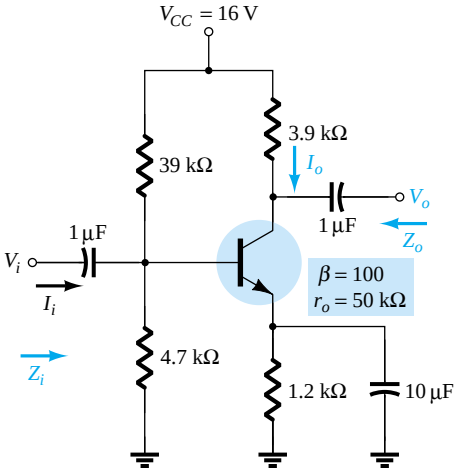


Figure 8.67 Problem 4

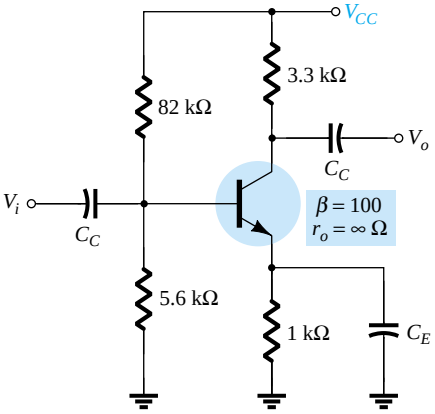


Figure 8.68 Problem 5

5. Determine V_{CC} for the network of Fig. 8.68 if $A_v = -160$ and $r_o = 100\ \text{k}\Omega$.

6. For the network of Fig. 8.69:
(a) Determine r_e.
(b) Calculate V_B and V_C.
(c) Determine Z_i and $A_v = V_o/V_i$.

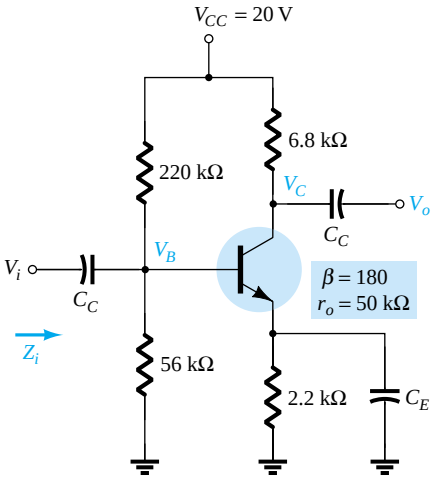


Figure 8.69 Problem 6

§ 8.4 CE Emitter-Bias Configuration

7. For the network of Fig. 8.70:
(a) Determine r_e.
(b) Find Z_i and Z_o.
(c) Calculate A_v and A_i.
(d) Repeat parts (b) and (c) with $r_o = 20\ \text{k}\Omega$.

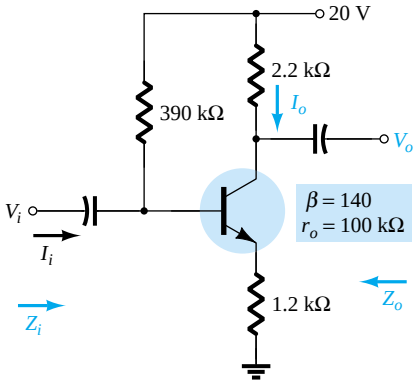


Figure 8.70 Problems 7 and 9

8. For the network of Fig. 8.71, determine R_E and R_B if $A_v = -10$ and $r_e = 3.8\ \Omega$. Assume that $Z_b = \beta R_E$.

9. Repeat Problem 7 with R_E bypassed. Compare results.

* 10. For the network of Fig. 8.72:
   (a) Determine r_e.
   (b) Find Z_i and A_v.
   (c) Calculate A_i.

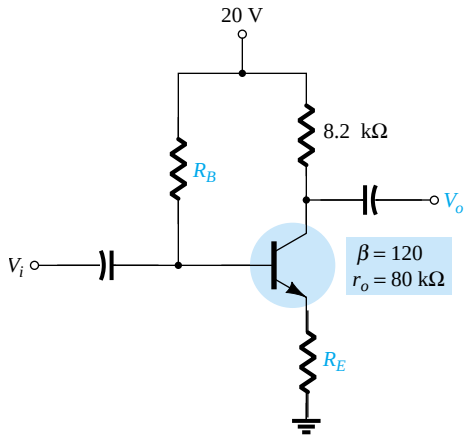


Figure 8.71 Problem 8

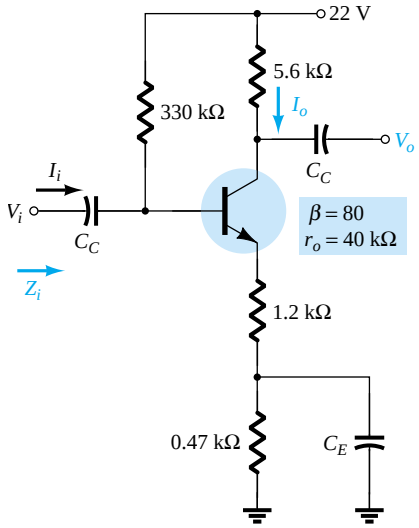


Figure 8.72 Problem 10

§ 8.5 Emitter-Follower Configuration

11. For the network of Fig. 8.73:
   (a) Determine r_e and βr_e.
   (b) Find Z_i and Z_o.
   (c) Calculate A_v and A_i.

* 12. For the network of Fig. 8.74:
   (a) Determine Z_i and Z_o.
   (b) Find A_v.
   (c) Calculate V_o if $V_i = 1$ mV.

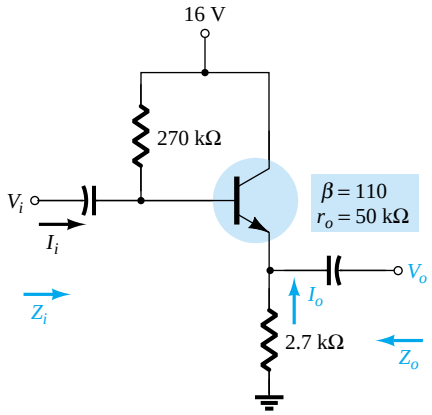


Figure 8.73 Problem 11

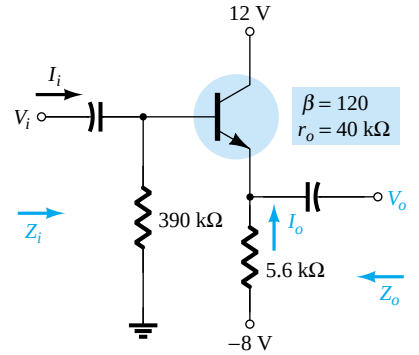


Figure 8.74 Problem 12

* **13.** For the network of Fig. 8.75:
(a) Calculate I_B and I_C.
(b) Determine r_e.
(c) Determine Z_i and Z_o.
(d) Find A_v and A_i.

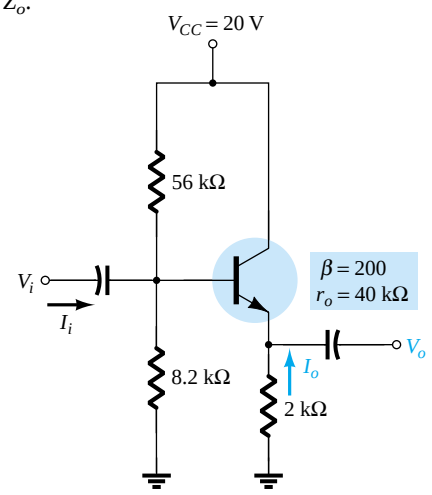


Figure 8.75 Problem 13

§ 8.6 Common-Base Configuration

14. For the common-base configuration of Fig. 8.76:
(a) Determine r_e.
(b) Find Z_i and Z_o.
(c) Calculate A_v and A_i.

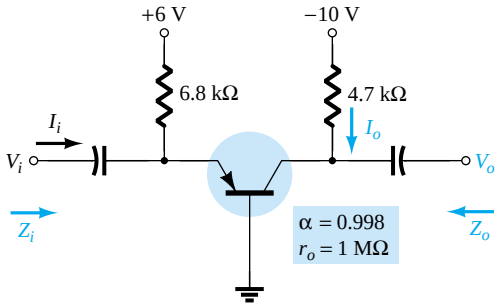


Figure 8.76 Problem 14

* **15.** For the network of Fig. 8.77, determine A_v and A_i.

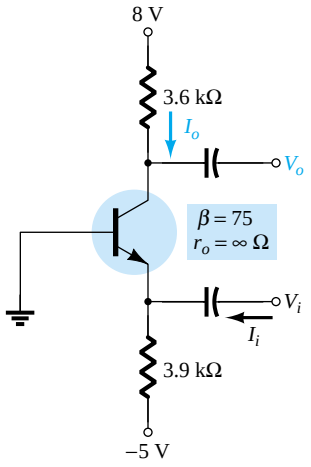


Figure 8.77 Problem 15

§ 8.7 Collector Feedback Configuration

16. For the collector FB configuration of Fig. 8.78:
(a) Determine r_e.
(b) Find Z_i and Z_o.
(c) Calculate A_v and A_i.

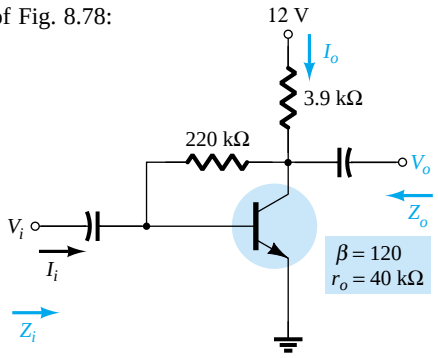


Figure 8.78 Problem 16

* **17.** Given $r_e = 10\ \Omega$, $\beta = 200$, $A_v = -160$, and $A_i = 19$ for the network of Fig. 8.79, determine R_C, R_F, and V_{CC}.

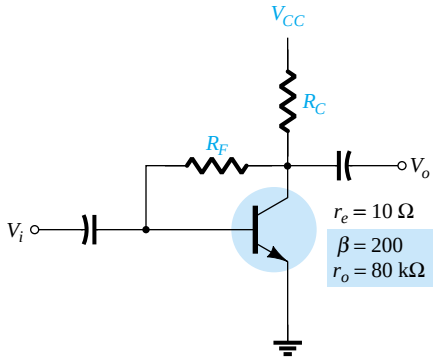


Figure 8.79 Problem 17

* **18.** For the network of Fig. 8.30:
(a) Derive the approximate equation for A_v.
(b) Derive the approximate equation for A_i.
(c) Derive the approximate equations for Z_i and Z_o.
(d) Given $R_C = 2.2$ kΩ, $R_F = 120$ kΩ, $R_E = 1.2$ kΩ, $\beta = 90$, and $V_{CC} = 10$ V, calculate the magnitudes of A_v, A_i, Z_i, and Z_o using the equations of parts (a) through (c).

§ 8.8 Collector DC Feedback Configuration

19. For the network of Fig. 8.80:
(a) Determine Z_i and Z_o.
(b) Find A_v and A_i.

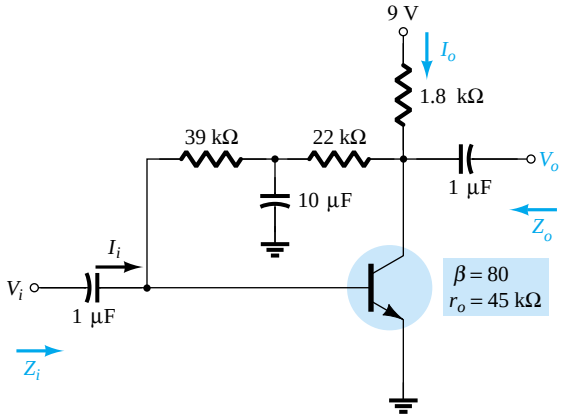


Figure 8.80 Problem 19

§ 8.9 Approximate Hybrid Equivalent Circuit

20. (a) Given $\beta = 120$, $r_e = 4.5\ \Omega$, and $r_o = 40\ \text{k}\Omega$, sketch the approximate hybrid equivalent circuit.
(b) Given $h_{ie} = 1\ \text{k}\Omega$, $h_{re} = 2\times10^{-4}$, $h_{fe} = 90$, and $h_{oe} = 20\ \mu\text{S}$, sketch the r_e model.

21. For the network of Problem 1:
(a) Determine r_e.
(b) Find h_{fe} and h_{ie}.
(c) Find Z_i and Z_o using the hybrid parameters.
(d) Calculate A_v and A_i using the hybrid parameters.
(e) Determine Z_i and Z_o if $h_{oe} = 50\ \mu\text{S}$.
(f) Determine A_v and A_i if $h_{oe} = 50\ \mu\text{S}$.
(g) Compare the solutions above with those of Problem 1. (Note: The solutions are available in Appendix E if Problem 1 was not performed.)

22. For the network of Fig. 8.81:
(a) Determine Z_i and Z_o.
(b) Calculate A_v and A_i.
(c) Determine r_e and compare βr_e to h_{ie}.

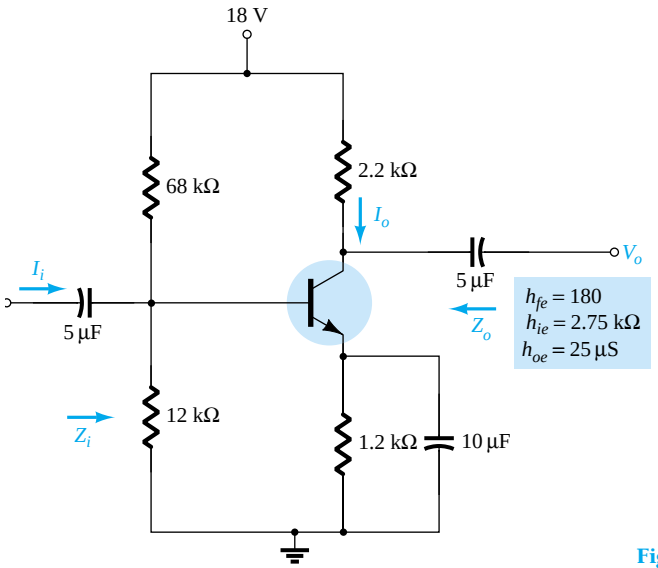


Figure 8.81 Problems 22 and 24

* **23.** For the common-base network of Fig. 8.82:
(a) Determine Z_i and Z_o.
(b) Calculate A_v and A_i.
(c) Determine α, β, r_e, and r_o.

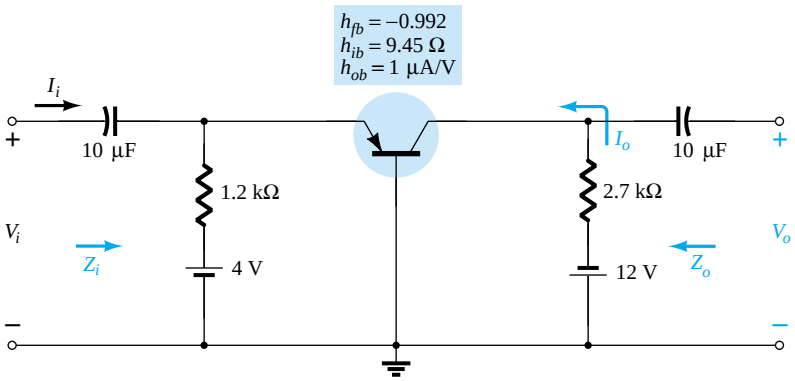


Figure 8.82 Problem 23

§ 8.10 Complete Hybrid Equivalent Model

* **24.** Repeat parts (a) and (b) of Problem 22 with $h_{re} = 2 \times 10^{-4}$ and compare results.

* **25.** For the network of Fig. 8.83, determine:
  (a) Z_i.
  (b) A_v.
  (c) $A_i = I_o/I_i$.
  (d) Z_o.

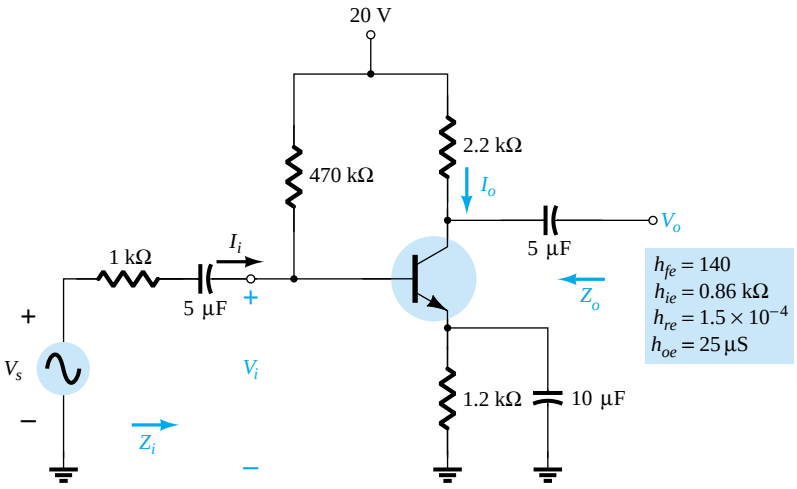


Figure 8.83 Problem 25

* **26.** For the common-base amplifier of Fig. 8.84, determine:
  (a) Z_i.
  (b) A_i.
  (c) A_v.
  (d) Z_o.

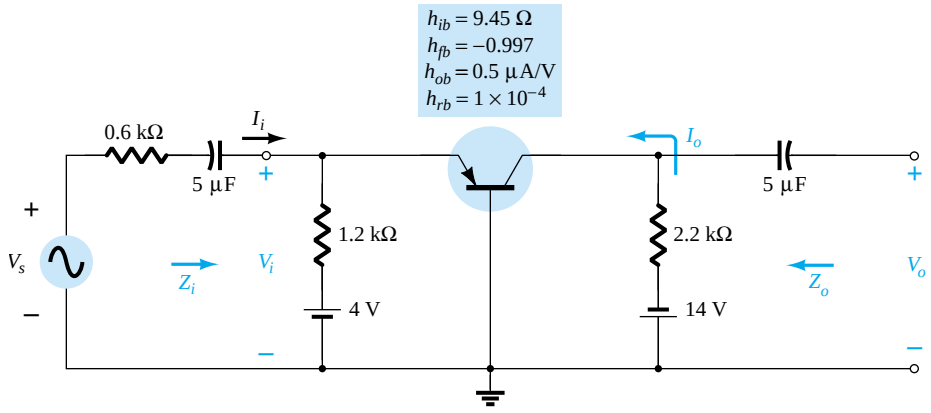


Figure 8.84 Problem 26

§ 8.12 Troubleshooting

* **27.** Given the network of Fig. 8.85:
  (a) Determine if the system is operating properly based on the voltage-divider bias levels and expected waveforms for v_o and v_E.
  (b) Determine the reason for the dc levels obtained and why the waveform for v_o was obtained.

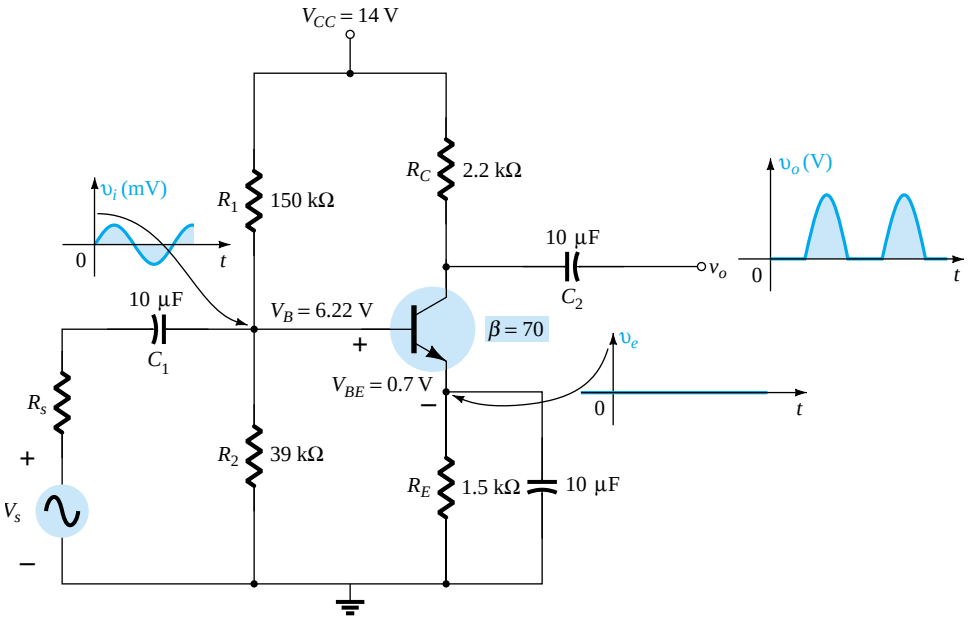


Figure 8.85 Problem 27

§ 8.13 PSpice Windows

28. Using PSpice Windows, determine the gain for the network of Fig. 8.6. Use Probe to display the input and output waveforms.

29. Using PSpice Windows, determine the gain for the network of Fig. 8.13. Use Probe to display the input and output waveforms.

30. Using PSpice Windows, determine the gain for the network of Fig. 8.25. Use Probe to display the input and output waveforms.

*Please Note: Asterisks indicate more difficult problems.